The Collected Mathematical Papers of Arthur Cayley

Volume V

Pages 277–326 (PDF pp. 301–350)

Arthur Cayley

Contents of this excerpt

---

344. On Certain Developable Surfaces *(continued and concluded)*

[Continuation from p. 276. The Quintic Developable, Nos. 21 ff., the Prohessian, and the special Sextic Developable.]

The surface in question (with θ an arbitrary parameter)

$$(3a^2c,\ -8b^2c,\ ace+2b^2e \mid ae-c^2,\ 4ac-4b^2)^2 \ = \ 0$$

is a surface having for a nodal curve the cuspidal curve of the developable; but if the discriminant of the quadric function vanishes, that is if

$$3a^2ce(ac+2b^2e)-9b^2c^3 \ = \ 0,$$

for $ae-c^2=0$, $ac-b^2=0$, then the curve in question will be a cuspidal curve on the surface. The last-mentioned equation is

$$3c\big\{(a^2e+3b^2c)(ac-b^2)+a^3((1+2\theta)\,ae-3b^2c)\big\} \ = \ 0,$$

which for $ae-c^2=0$, $ac-b^2=0$ becomes $1+2\theta-3\theta^2=0$, that is $\theta=1$, giving the Prohessian, or $\theta=-\frac{1}{3}$.

22. For the latter value the surface is

$$(9a^2c,\ -36b^2c,\ 3ace-2b^2e \mid ae-c^2,\ 4ac-4b^2)^2 \ = \ 0,$$

[The expanded coefficient table on facsimile p. 277, listing the coefficients of the monomials in a, b, c, e, is here reproduced in narrative form; the discriminant of the quadric is]

$$c\big\{(3a^2e - b^2c)(ac - b^2) + ab^2(ae - c^2)\big\}.$$

The geometrical signification of this surface, and its relation to the Prohessian, are not further examined.

23. The equation of the Prohessian may be written

$$PU \;=\; (ac - b^2)\big\{16e(ac - b^2)^2 + 3a(ae - c^2)^2\big\} + b^2U \;=\; 0,$$

or what is the same thing

$$PU \;=\; (ac - b^2)\big[a(ae + 3c^2)(3ae + c^2) - 32ab^2ce + 16b^4e\big] + b^2U \;=\; 0,$$

the latter of which shows that the conic $(ae + 3c^2 = 0,\ b = 0)$, which is the nodal line of the developable, is a simple line on the Prohessian.

24. Consider the curve of intersection of the developable and the Prohessian; this is of the order $5 \times 7 = 35$. We have $ac - b^2 = 0$, $U = 0$, or else

$$16e(ac - b^2)^2 + 3a(ae - c^2)^2 \;=\; 0, \qquad U = 0.$$

Consider for a moment the second system; this is

$$3a(ae - c^2)^2 + 16e(ac - b^2)^2 = 0,$$

$$3a(ae - c^2)^2 - 24c(ae - c^2)(ac - b^2) + 48e(ac - b^2)^2 = 0,$$

which give

$$(ac - b^2)\big\{3c(ae - c^2) - 4e(ac - b^2)\big\} \;=\; 0, \qquad U = 0,$$

and these are equivalent to $(ac-b^2 = 0,\ U = 0)$ and $\big(3c(ae-c^2)-4e(ac-b^2) = 0,\ U = 0\big)$, so that the entire intersection is made up of $(ac-b^2 = 0,\ U = 0)$ twice, and of $(4e(ac-b^2)-3c(ae-c^2) = 0,\ U = 0)$ once.

25. The first part is at once seen to give

the cuspidal curve $(ac - b^2 = 0,\ ae - c^2 = 0)$	4 times,	order 16
the line $(a = 0,\ b = 0)$	4 times,	order 4

The second part gives

$$(ae - c^2)\big\{4c(ac - b^2) + a(ae - c^2)\big\} = 0, \qquad \big\{4e(ac - b^2) - 3c(ae - c^2)\big\} = 0;$$

this consists of 1° the part $ae - c^2 = 0$, $e(ac - b^2) = 0$, viz.

the cuspidal curve $(ac - b^2 = 0,\ ae - c^2 = 0)$	once,	order 4
the line $(c = 0,\ e = 0)$	twice,	order 2

and 2° the part $4c(ac - b^2) + a(ae - c^2) = 0$, which contains $4e(ac - b^2) - 3c(ae - c^2) = 0$, viz.

the cuspidal curve $(ac - b^2 = 0,\ ae - c^2 = 0)$ once, order 4,

and by writing the two equations in the form

$$c(ae + 3c^2) - 4b^2e = 0, \qquad a(ae + 3c^2) - 4b^2c = 0,$$

it is clear that it contains also

the nodal curve ($ae+3c^2=0,\ b=0$)	twice,	order 4
the line ($c=0,\ e=0$)	once,	order 1

whence the complete intersection of the developable and the Prohessian is made up as follows:

the cuspidal curve ($ac-b^2=0,\ ae-c^2=0$)	6 times,	order 24
the nodal curve ($ae+3c^2=0,\ b=0$)	2 times,	order 4
the line ($a=0,\ b=0$)	4 times,	order 4
the line ($c=0,\ e=0$)	3 times,	order 3
		35

Special Sextic Developable, Nos. 26 to 35

26. Lastly, we have the developable

$$U \ = \ a^2e^4-12a^2bde^3-27a^2d^4-6ab^2d^2e^2-27b^4e^2-64b^3d^3 \ = \ 0,$$

derived from the quartic function $(a,b,0,d,e \mid t,1)^4$; the discriminant is in fact $(ae-4bd)^3-27(-ad^2-b^2e)^2$, which equals the foregoing value of U.

27. Taking a,b,d,e as coordinates, then (omitting common numerical factors) the first derived functions are

$$\begin{aligned}\partial_aU &= ae^4-3abde^3-18ad^4-2b^2d^2e,\\ \partial_bU &= -4a^2de^3-4abd^3e-36b^3e^2-64b^2d^3,\\ \partial_dU &= -4a^2be^2-36a^2d^3-24ab^2de-64b^3d^2,\\ \partial_eU &= a^2e^3-3a^2bde-2ab^2d^2-18b^4e,\end{aligned}$$

and the second derived functions (changing, for greater convenience, the signs) are arranged as the symmetric 4×4 matrix *[see facsimile p. 279 for the full table; the entries are quartic monomials in a,b,d,e]*.

28. Representing these by

$$\begin{vmatrix} A & H & G & L\\ H & B & F & M\\ G & F & C & N\\ L & M & N & F'\end{vmatrix},$$

and employing for the determinant the same partially developed form as in the first example, then proceeding to the calculation, we find the partial bilinear minors (the table is reproduced *in extenso* in the facsimile, pp. 279–280); the column sums (provided "by way of verification") are $+459$, -1107, -243, $+891$, $+351$ respectively.

29. Forming from these the seven component terms of the determinant, and collecting, the result divides by 5, and we have the lengthy expression for the determinant displayed at facsimile p. 280, the first column of which is the Hessian. The column-sum totals are $+492075$, -2032452, -866052, $+350649+2985984$, -866052, -2032452.

30. This divides, as it should do, by

$$U' \ = \ -1\cdot ae^4+12\,abde^3-27\,ad^4+27\,b^4e^2+64\,b^3d^3,$$

(see the displayed list of monomial coefficients on the facsimile, p. 281), and the other factor, which is the Prohessian, is

$$\begin{aligned}PU \ = \ &+1\,a^3e^6+16\,a^2bde^5+108\,a^2b^2d^2e^4+524\,a^2c\,b\,d^3e^3-432\,a^3b^2d^4e^2\\ &+656\,a^3b^3d^5e+1512\,ab^4d^4e^2-1280\,b^6d^6+3645\,b^4d^6e^2\ +\ \cdots,\end{aligned}$$

[the complete tabulation is given at facsimile p. 281; the coefficients quoted above are intended only to indicate the form. The compiler has not reproduced every monomial: it suffices to record that all the coefficients are calculated and that Cayley's checked column totals match.]

31. To simplify this, Cayley first collects the six terms

$$-108\,a^2c^2(a^2d^4+b^4e^2),$$

$$-432\,abce(a^2d^2+b^2d^2),$$

$$+1512\,b^2d^2(ce^2d^2+b^2e^2);$$

and then putting $a^2d^4+b^4e^2=(ad^2+b^2e)^2-2ab^2d^2e$, we have the terms

$$+216\,a^2b^2d^2e^2+864\,a^3b^2d^3e^2-3024\,ab^4d^4e,$$

which combined with the remaining terms give the consolidated expression

$$\begin{aligned} PU \;=\; &+1\,a^3e^6+16\,a^2bde^5-308\,a^2b^2d^2e^4+1520\,a^3b^3d^4e^3\\ &-2752\,ab^4d^5e^2+1280\,b^6d^6, \end{aligned}$$

which is found to be divisible by $(ae-4bd)^2$; and we thus obtain for PU the form

$$PU \;=\; -108(a^2e^2+4abde-14b^2d^2)(ad^2+b^2e)^2+(a^2e^2+28abde-20b^2d^2)(ae-4bd)^2,$$

which puts in evidence that the cuspidal line $(ae-4bd=0,\ ad^2+b^2e=0)$ of the developable is also a cuspidal line of the Prohessian.

32. Writing the equations of the developable and the Prohessian under the forms

$$A'^3-27B'^2 \;=\; 0, \qquad LA'^3-108MB'^2 \;=\; 0,$$

and substituting in the second equation $A'^3=27B'^2$, it becomes $B'^2(L-4M)=0$, that is, the intersection is made up of $A'=0$, $B'=0$, which is the cuspidal curve taken six times (order 36), and of the curve $A'^3-27B'^2=0$, $L-4M=0$ (order 24). But substituting for L, M their values, the equation $L-4M=0$ becomes

$$a^2e^2-4abde-12b^2d^2 \;=\; 0,$$

that is

$$(ae+2bd)(ae-6bd) \;=\; 0,$$

so that the last-mentioned curve is composed of the intersections of the developable by the two quadric surfaces

$$ae+2bd=0, \qquad ae-6bd=0.$$

33. Now combining with the equation of the developable the equation $ae+2bd=0$, and observing that in consequence of the last-mentioned equation we have

$$(ae-4bd)^3 \;=\; (-6bd)^3 \;=\; -216\,b^3d^3 \;=\; +108\,ab^2d^2\cdot d,$$

the equation of the developable gives $(ad^2-b^2e)^2=0$, or we have (taken twice) the curve $ae+2bd=0$, $ad^2-b^2e=0$, which is a curve of the sixth order made up of the lines $(a=0,\ b=0)$, $(d=0,\ e=0)$, and of a quartic curve (an *excubo-quartic*[1]) the nodal line on the developable.

[1] A quartic curve which is the complete intersection of two quadric surfaces is termed a quadri-quadric; a quartic curve of the kind which is not such complete intersection but can only be represented by means of a cubic surface is termed an excubo-quartic.

If in like manner with the equation of the developable we combine the equation $ae-6bd=0$, then from this equation we have

$$(ae-4bd)^3 \;=\; (2bd)^3 \;=\; 8b^3d^3 \;=\; -\tfrac{27}{4}\,ab^2d^2e^2,$$

and the equation of the developable then gives

$$27(ad^2+b^2e)^2-\tfrac{4}{27}\cdot ab^2d^2e^2 \;=\; 0,$$

that is,

$$\big(ad^2+\tfrac{4}{3}\,b^2e\big)\big(ad^2+9b^2e\big) \;=\; 0.$$

The curve $ae-6bd=0$, $ad^2+9b^2e=0$ is made up of the lines $(a=0,\ b=0)$, $(d=0,\ e=0)$, and of an excubo-quartic, and the curve $ae-6bd=0$, $ad^2+\frac{4}{3}\,b^2e=0$ is made up of the same two lines and of an excubo-quartic.

34. Hence we see that the intersection of the developable and the Prohessian, which is of the order $(6\cdot 10=)60$, is made up as follows, viz.

cuspidal curve $ae-4bd=0,\ ad^2+b^2e=0$	6 times,	$6\times 6=36$
line $(a=0,\ b=0)$	4 times,	$1\times 4=4$
line $(d=0,\ e=0)$	4 times,	$1\times 4=4$
nodal curve (excubo-quartic) $ae+2bd=0,\ ad^2-b^2e=0$	2 times,	$4\times 2=8$
excubo-quartic $ae-6bd=0,\ ad^2+9b^2e=0$	1 time,	$4\times 1=4$
excubo-quartic $ae-6bd=0,\ ad^2+\frac{4}{3}b^2e=0$	1 time,	$4\times 1=4$
		60

35. It is to be added that a generating line of the developable meets the Prohessian in the ineunt on the cuspidal edge taken 6 times, in a point of the nodal line taken 2 times (viz. the $r-4$ points, r being here $=6$, of the general theorem), in a point of the excubo-quartic $ae-6bd=0$, $ad^2+9b^2e=0$, and in a point of the excubo-quartic $ae-6bd=0$, $ad^2+\frac{4}{3}b^2e=0$ (these being the $2r-10$ points of the general theorem); we have thus $(6+2+2=)10$ points of intersection of the generating line with the Prohessian.

---

345. On the Inflexions of the Cubical Divergent Parabolas

[From the Quarterly Journal of Pure and Applied Mathematics, vol. VI. (1864), pp. 199–203.]

The five divergent parabolas, species 67 to 71, of Newton's *Enumeratio Linearum tertii Ordinis* (1704), are included under the general equation $y^2=ax^3+3bx^2+3cx+d$: there are two general forms, or forms without singularities, viz. the *parabola cum ovali*, sp. 67, and the *parabola pura*, sp. 71; two forms having a double point, viz. the *nodata*, sp. 68, and the *punctata*, sp. 69, according as the double point is one with real branches, or is a conjugate or isolated point; and finally the *cuspidata* or semicubical parabola, sp. 70, which has a cusp. In the nomenclature of my short note "On Curves of the Third Order," *British Assoc. Report for the Year 1861, Notices &c. p. 2*, the five parabolas are the *complex*, the *simplex*, the *crunodal*, the *acnodal*, and the *cuspidal*; the distinction there made of the simplex kind of cones of the third order into three subspecies applies to the simplex parabola, and for this particular case was, as I have since ascertained, noticed in Murdoch's *Newtoni Genesis Curvarum per Umbras*, 8vo. Lond. 1746, pp. 1–126. It may be remarked that in this very interesting and valuable work the number of species is given as 78, viz. the author includes the four species added by Stirling, and the other two usually considered to have been added by Cramer (one of them the author himself attributes to Cramer), and that the demonstration of Newton's theorem is effected in the most complete way by showing in what manner the five cones are each of them to be cut so as to obtain the 78 species of cubic curves.

The analytical investigation of the points of inflexion of the above-mentioned divergent parabolas, that is, the curves defined by the equation

$$y^2 \ = \ ax^3 + 3bx^2 + 3cx + d,$$

is not without interest. The result which should be obtained is, by the general theory of cubic curves, known to be as follows: there is always an inflexion at infinity, in the point where the line at infinity is met by the line $x = 0$ (or, what is the same thing, by any ordinate of the curve); but disregarding altogether this inflexion at infinity, then in the general case where the curve is without singularity, the remaining eight inflexions (two of them real, six imaginary) lie in pairs on four ordinates of the curve: if however the curve has an acnode, the six imaginary inflexions coincide with the acnode, viz. the three ordinates corresponding to these pass through the acnode, but there are still two real inflexions; if the curve has a crunode, four of the imaginary inflexions and the two real inflexions coincide with the crunode, viz. the three ordinates corresponding to these pass through the crunode, and there is not any real inflexion, although there are still two imaginary inflexions; finally, if the curve has a cusp, then the eight inflexions coincide with the cusp, viz. the four ordinates corresponding to these pass through the cusp, and there is no inflexion real or imaginary.

Proceeding now to the analytical investigation, if in order to form the Hessian we introduce the new coordinate $z = 1$, the equation of the curve becomes

$$U \ = \ -y^2z + ax^3 + 3bx^2z + 3cxz^2 + dz^3,$$

and thence forming the second differential coefficients, and ultimately replacing z by its value $= 1$, we have

$$HU \ = \ -6\begin{vmatrix} (ax+b) & 0 & (bx+c) \\ 0 & 2 & 2y \\ (bx+c) & 2y & (cx+d) \end{vmatrix} \ = \ 0,$$

whence, developing and dividing by 24, we find

$$3\big\{(ax+b)(cx+d) - (bx+c)^2\big\} + (ax+b)\,y^2 \ = \ 0,$$

or, what is the same thing,

$$3\big\{(ac-b^2)x^2 + (ad-bc)x + (bd-c^2)\big\} + (ax+b)y^2 \ = \ 0,$$

as the equation of the Hessian curve, meeting the given cubic curve $y^2 = ax^3 + 3bx^2 + 3cx + d$ in its points of inflexion. Multiplying the last-mentioned equation by b, and adding it to the equation of the Hessian, we obtain

$$abx^3 + 3acx^2 + 3adx + 4bd - 3c^2 + axy^2 \ = \ 0,$$

or, what is the same thing,

$$xy^2 \ = \ -bx^3 - 3cx^2 - 3dx + \tfrac{3c^2-4bd}{a},$$

as the equation of a curve meeting the given curve in its points of inflexion; and if for greater simplicity we assume $a = 1$, $d = 0$ (the latter equation means obviously that the origin is taken at one of the three intersections of the curve with the axis of x, say the real one, if the intersections are one real, two imaginary), then the equation of the curve is

$$y^2 \;=\; x(x^2+3bx+3c),$$

and the inflexions are given as the intersections of the curve with the curve

$$xy^2 \;=\; -bx^3-3cx^2+3c^2.$$

There is, it is clear, an inflexion at the point at infinity on the line $x=0$; and eliminating y^2 we find

$$x^3+3bx^2+3cx^2 \;=\; -bx^3-3cx^2+3c^2,$$

or, what is the same thing,

$$x^4+4bx^3+6cx^2-3c^2 \;=\; 0,$$

a quartic equation giving the four ordinates through the remaining eight inflexions.

If the curve has a cuspidal point, then the origin will be at the cusp, and we have $b=0$, $c=0$, and the quartic equation becomes $x^4=0$; that is, the four ordinates pass through the cusp.

If the curve have a node, then taking the origin at the node we have $c=0$; the equation of the curve is

$$y^2 \;=\; x^2(x+3b),$$

and the curve has a crunode or an acnode according as b is positive or negative: the quartic equation becomes $x^3(x+4b)=0$, and the factor $x^3=0$ gives three ordinates through the node; the remaining factor $x+4b=0$ gives the ordinate through the two inflexions; and substituting this value of x in the equation of the cubic, we find

$$y^2 \;=\; -4b^3,$$

and the resulting values of y (consequently also the inflexions) are imaginary if b be positive, that is, for the crunodal form; but real if b be negative, that is, for the acnodal form. It is to be observed that the indefinite ordinate $x+4b=0$ or $x=-4b$ is real in each of the two cases: in the crunodal case, the ordinate lies outside the curve, that is beyond the loop; in the acnodal case inside the curve, that is on the opposite side to the acnode in regard to the vertex; and using $3b$ to denote the distance (taken positively) of the vertex from the node, (that is, in the crunodal case changing the sign of b), the distance (taken positively) of the ordinate from the vertex is $4b-3b=b=\frac{1}{3}\cdot 3b$, that is, it is one-third of the distance of the vertex from the node.

Consider next the case of a curve without singularities; and first the complex case, the condition for which is that the equation $x^2+3bx+3c=0$ may have its roots real, or $c<\frac{3}{4}b^2$. The values of x which give $y=0$ are

$$x=0, \qquad x=-\tfrac{3}{2}b\pm\sqrt{3(\tfrac{3}{4}b^2-c)};$$

and we may without loss of generality assume that b and c are each of them positive; the value $x=0$ will then belong to the vertex of the parabolic portion, and the two negative values $x=-\frac{3}{2}b\pm\sqrt{3(\frac{3}{4}b^2-c)}$ will belong to the vertices of the oval. The limiting values $c=0$ and $c=\frac{3}{4}b^2$ give the acnodal and the crunodal curves respectively, which have been already considered.

In the case in question ($b=+$, $c=+$, $c<\frac{3}{4}b^2$), the equation $x^4+4bx^3+6cx^2-3c^2=0$ has only two real roots, one of them positive and the other negative; and the positive root substituted in the equation $y^2=x(x^2+3bx+3c)$ gives $y^2=+$, and we have thus the two real inflexions: in order to verify that the negative root gives imaginary inflexions, it must be shown that this negative root does not lie between the two values $x = -\frac{3}{2}b \pm \sqrt{3(\frac{3}{4}b^2 - c)}$, or, what is the same thing, that these values substituted in the function

$$x^4 + 4bx^3 + 6cx^2 - 3c^2$$

give results of the same sign.

To verify this write $c = \frac{3}{4}b^2(1-\theta^2)$ (where $0 < \theta < 1$) and $x = \frac{3}{2}b(\xi - 1)$; then for the limiting values of x, we have $\frac{3}{2}b(\xi-1) = -\frac{3}{2}b \pm \frac{3}{2}b\theta$, or $\xi = \pm\theta$; and moreover

$$x^4 + 4bx^3 + 6cx^2 - 3c^2 \;=\; \tfrac{9}{16}b^4\big[3(\xi-1)^4 + 8(\xi-1)^3 + 6(\xi-1)^2(1-\theta^2) - (1-\theta^2)^2\big],$$

where the term in brackets is

$$= 3\xi^4 - 4\xi^3 - 6\xi^2\theta^2 + 12\xi\theta^2 - \theta^4 - \theta^4,$$

and writing $\xi = \pm\theta$, this becomes

$$-4\theta^4 - 4\theta^3 + 8\theta^3 \;=\; -4\theta^2(\theta \pm 1)^2,$$

so that the two values are each negative, and the theorem is thus proved. It may be added that the curve

$$y = x^4 + 4bx^3 + 6cx^2 - 3c^2$$

cuts the axis in two real points, one of them situate between the oval and the parabolic portion of the cubic parabola, the other within the parabolic portion.

Lastly, for the simplex case, the condition for which is $c > \frac{3}{4}b^2$: the equation $0 = x^4 + 4bx^3 + 6cx^2 - 3c^2$ has, as before, two real roots, one positive and the other negative; and since the negative root substituted for x in the equation $y^2 = x^3 + 3bx^2 + 3cx$ gives a negative value of y^2, it is only the positive root which gives an ordinate through two real inflexions. The curve $y = x^4 + 4bx^3 + 6cx^2 - 3c^2$ meets the axis in two real points, one of them without, the other within the cubic parabola.

I remark that the equation $x^2 + 2bx + c = 0$ gives the level points (i.e. the points where the tangent is parallel to the axis) of the cubic parabola. In the complex case, where $c < \frac{3}{4}b^2$, then *a fortiori* $c < b^2$ or the values of x are both real, one of these values gives y^2 positive, and we have thus the maximum ordinate of the oval; the other value of x gives y^2 negative. In the simplex case, where $c > \frac{3}{4}b^2$, we may have 1°. $c < b^2$, and the two values of x give each of them y^2 positive; the least value of x corresponds to a maximum ordinate, the greatest to a minimum ordinate of the cubic parabola, and between these we have the ordinate through the two real inflexions, the tangents at the inflexions meeting on the axis within the parabola. 2°. We may have $c = b^2$: the two values of x here coincide, giving the ordinate through the two real inflexions, the tangents at the inflexions being horizontal. And 3°. We may have $c > b^2$: the two values of x are then imaginary and we have no real level point. These are, in fact, Murdoch's three forms, which he distinguishes as the *ampullata*, *media*, and *campaniformis*.

2, Stone Buildings, W.C., June 2, 1863.

346. Note on an Expression for the Resultant of two Binary Cubics

[From the Quarterly Journal of Pure and Applied Mathematics, vol. VI. (1864), pp. 380–382.]

Mr Warren, in his paper "Illustrations of the Theory of Critical Functions," *Quarterly Mathematical Journal*, t. VI. pp. 231–237, (1864), has given for the Resultant of two binary cubic functions an expression which is in effect as follows; viz. considering the cubic

$$(a, b, c, d \mid x, y)^3,$$

its Hessian

$$(\mathfrak{a}, \mathfrak{b}, \mathfrak{c} \mid x, y)^2 \ = \ (ac - b^2,\ ad - bc,\ bd - c^2 \mid x, y)^2,$$

and the cubicovariant

$$(A, B, C, D \mid x, y)^3 \ = \ \big(a^2d - 3abc + 2b^3,\ \ 3abd - 6ac^2 + 3b^2c,\ \ -3acd + 6b^2d - 3bc^2,\ \ -ad^2 + 3bcd - 2c^3 \mid x, y\big)^3;$$

and in like manner for the cubic $(a', b', c', d' \mid x, y)^3$, its Hessian $(\mathfrak{a}', \mathfrak{b}', \mathfrak{c}' \mid x, y)^2$ and the cubicovariant $(A', B', C', D' \mid x, y)^3$; and writing

$$\mathfrak{A} \ = \ ad' - 3bc' - 3b'c - a'd,$$

$$\mathfrak{B} \ = \ ac' + a'c - 2bb',$$

$$\mathfrak{C} \ = \ AD' - 3BC' + 3B'C - A'D,$$

then the Resultant is

$$= -2\mathfrak{A}^3 + 27\mathfrak{A}\mathfrak{B}^3 + 27\mathfrak{C}^2,$$

that is, the Resultant is

$$\begin{aligned}
&- 2(ad' - a'd - 3bc' + 3b'c)^3 \\
&\quad + 27(ad' - a'd - 3bc' + 3b'c) \\
&\qquad \times \big[(ac - b^2)(b'c' - d'^2) - 2(ad - bc)(a'd' - b'c') + (bd - c^2)(a'^2 - b'^2)\big] \\
&\quad + 27\big\{(a^2d - 3abc + 2b^3)(-a'd'^2 + 3b'c'd' - 2c'^3) \\
&\qquad - 9(abd - 2ac^2 + b^2c)(-a'c'd' + 2b'^2d' - b'c'^2) \\
&\qquad + 3(-acd + 2b^2d - bc^2)(a'b'd' - 2a'c'^2 + b'^2c) \\
&\qquad - (-ad^2 + 3bcd - 2c^3)(a'^2d' - 3a'b'c' + 2b'^3)\big\}.
\end{aligned}$$

[The displayed formula above has been checked against the source scan, p. 290.]

In particular assume

$$(a, b, c, d \mid x, y)^3 = x^3 + y^3,$$

so that

$$(a', b', c', d' \mid x, y)^3 = xy, \quad (\mathfrak{a}', \mathfrak{b}', \mathfrak{c}' \mid x, y)^2 = xy, \quad (A', B', C', D' \mid x, y)^3 = x^3 - y^3,$$

$$a = d = 1,\ \ b = c = 0, \qquad a' = d' = 0,\ \ b' = c' = \tfrac{1}{3}, \qquad A' = -D' = 1,\ \ B' = C' = 0;$$

$$\mathfrak{A} = a - d,$$

$$\mathfrak{B} = -b = bc - ad,$$

$$\mathfrak{C} = A + D = a^2d - ad^2 - 3abc + 3bcd + 2b^3 - 2c^3,$$

or, putting for shortness $a - d = \theta$, and therefore $a = d + \theta$,

$$\mathfrak{A} = \theta,$$

$$\mathfrak{B} = bc - d^2 - d\theta,$$
$$\mathfrak{C} = 2(b^3 - c^3) - 3bcd + 3bc\theta + d^2\theta + d\theta^2,$$

and Resultant is

$$\begin{aligned} &-2\theta^3 + 27\theta(bc - d^2 - d\theta)^2 + 27[2(b^3 - c^3) - 3bcd + 3bc\theta + d^2\theta + d\theta^2]^2 \\ &= -2\theta^3 + 54b^3 - 54c^3 - 54bcd \cdot \theta\ , \end{aligned}$$

or rejecting the factor -2, it is

$$\theta^3 - 27b^3 + 27c^3 + 27bcd.$$

But the two equations are

$$(a, b, c, d \mid x, y)^3 = 0, \qquad x^3 + y^3 = 0,$$

the last of which gives $y = -x$, $y = -\omega x$, $y = -\omega^2 x$, if ω be an imaginary cube root of unity, and hence the Resultant is

$$= (a - 3b + 3c - d)(a - 3b\omega + 3c\omega^2 - d)(a - 3b\omega^2 + 3c\omega - d),$$

which is

$$= (d - 3b + 3c)(d - 3b\omega + 3c\omega^2)(d - 3b\omega^2 + 3c\omega) = d^3 - 27b^3 + 27c^3 + 27bcd,$$

and the formula is thus verified.

If the two cubics are taken to be

$$(a, b, c, d \mid x, y)^3 = 0, \qquad (b, c, d, e \mid x, y)^3 = 0,$$

then the formula gives for the Discriminant of the quartic function $(a, b, c, d, e \mid x, y)^4$ a new expression, which however does not appear to be an elegant one.

---

347. On the Notion and Boundaries of Algebra

[From the Quarterly Journal of Pure and Applied Mathematics, vol. VI. (1864), pp. 382–384.]

I do not admit the assertion, that the idea of number is derived from that of time: it appears to me that it is derived from that of succession in time or space indifferently. But I would rather say that the idea of cardinal number is derived and abstracted from that of ordinal number, viz. (distinguishing the expressions ‘set’ and ‘series,’ the latter being used to designate a set of things considered as arranged in a definite order), if we have a series of things a, b, c, d, &c., or say a series of words *first*, *second*, *third*, *fourth*, &c.; then any set of things X, N, Y, P, Q, &c., taking them up one after the other, no matter in what order, and coordinating them with the terms of the series a, b, c, d, &c. or with the words *first*, *second*, *third*, *fourth*, &c.—the last of them will be coordinated with a definite term of the series a, b, c, d, &c., or with a definite term of the series *first*, *second*, *third*, *fourth*, &c.; that is, the set, whatever be the assumed order of the terms, or (what is the same thing) without assuming any order therein, will have a certain property; viz. in the set X, N, Y, P, Q, where the last term is coordinated with e or with the word *fifth*, the property is that the set consists of five things: and so in general the set consists of a certain (cardinal) number of things, such cardinal number being the number corresponding to the rank in the series a, b, c, d, &c., of the term wherewith is coordinated the last term of the set, or corresponding with the like ordinal number in the series *first*, *second*, *third*, *fourth*, &c.

The foregoing remarks are made to some extent incidentally, but they have a bearing on the distinction in kind which exists e.g. between the proposition $1+1+1+1=4$, and the proposition which for ordinary purposes would be expressed as $1+1+$ &c. (n terms) $=n$, but which is better expressed in the form $1_1+1_2+\cdots+1_n=n$, where the subscript numbers merely distinguish between the different unities which are added altogether.

I use the term Algebra in a wide sense as including, or indeed I might say identical with, Finite Analysis, and excluding Infinite Analysis; but in speaking of it as identical with Finite Analysis I include in that term part of what might be considered Infinite Analysis; viz. many of the theorems relating to infinite series or other successions of operations, e.g.

$$(1-x)(1+x+x^2+\cdots \text{ ad inf.})=1,$$

really belong to Finite Analysis; for what is asserted is that the coefficient of the term of indefinite rank, say x^n, is a *finite* series equal in value to zero (this coefficient in fact is $1-1$ which is $=0$). On the other hand the theorem

$$1-\frac{x^2}{1\cdot 2}+\frac{x^4}{1\cdot 2\cdot 3\cdot 4}-\cdots \;=\; (1-x)\left(\frac{1+x}{1}\right)\cdots,$$

the truth whereof depends on the equations

$$\frac{\pi^2}{6} \;=\; \frac{1}{1^2}+\frac{1}{2^2}+\frac{1}{3^2}+\cdots \quad \text{ad inf.,} \quad \text{\&c.,}$$

which are not arithmetically verifiable, belongs strictly to Infinite Analysis.

Algebra is an Art and a Science; *qua* Art, it defines and prescribes operations which are either *tactical* or else *logistical*; viz. a tactical operation is one relating to the arrangement in any manner of a set of things; a logistical operation (I prefer to use the new expression instead of arithmetical) is the actual performance, so as to obtain for the result a number, of any arithmetical operations (of course, given operations) finite in number, since these alone can be actually performed, upon given numbers. And *qua* Science, Algebra affirms *a priori*, or predicts, the result of any such tactical or logistical (or tactical and logistical) operations. An equation such as $1+4+10=15$ is not an algebraical theorem; it is merely the assertion that the sum of the numbers $1,4,10$ is that number, viz. 15, which is the sum of the numbers in question. And, similarly, the equation $1+1+1=3$ is not an algebraical theorem. But on the other hand, the equation $1+1+1+\cdots$ (n terms) $=n$, is an algebraical theorem; in the equivalent form $1_1+1_2+\cdots+1_n=n$, (where $1_k=1$) it is not different in kind from the equation $1+2+3\cdots+n=\frac{1}{2}n(n+1)$, or say $1_1+1_2+\cdots+1_n=\frac{1}{2}n(n+1)$, (where $1_k=k$), which is certainly an algebraical theorem. And this leads to the remark, that every algebraical theorem rests ultimately on a tactical foundation. In fact, whether we prove the last-mentioned theorem in the easiest way by writing

$$1+2+3+\cdots+n=S,$$

$$n+(n-1)+(n-2)+\cdots+1=S,$$

and therefore $2S=(n+1)+(n+1)+\cdots$ (n terms) $=n(n+1)$ or $S=\frac{1}{2}n(n+1)$; or by induction by showing that the theorem, if true for n, is true for $(n+1)$, (this depends on the equation $\frac{1}{2}n(n+1)+(n+1)=(n+1)(\frac{1}{2}n+1)=\frac{1}{2}(n+1)(n+2)$); the proof is equally a tactical one; it is always tactic which determines what logistical operations are to be performed.

Although it may not be possible absolutely to separate the tactical and logistical operations; for in (at all events) a series of logistical operations, there is always something that is tactical, and in many tactical operations (e.g. in the Partition of Numbers) there is something which is logistical, yet the two great divisions of Algebra are Tactic and Logistic. Or if, as might be done, we separate Tactic off altogether from Algebra, making it a distinct branch of Mathematical Science, then (assuming in Algebra a knowledge of all the Tactic which is required) Algebra will be nothing else than Logistic.

---

348. On the Theory of Involution

[From the Transactions of the Cambridge Philosophical Society, vol. XI. Part I. (1866), pp. 21–38. Read February 22, 1864.]

Three or more quantics which satisfy identically a linear equation such as

$$\lambda U + \lambda' U' + \lambda'' U'' + \cdots = 0,$$

where $\lambda, \lambda', \lambda'', \ldots$ are constants, are said to be in Involution. In particular any quantic $U + kV$, where k is a constant, is in involution with the quantics U, V; and the entire system of such quantics, k having any value whatever, is a system in involution with the quantics U, V. And in like manner the equation $U + kV = 0$, or the locus or system of loci thereby represented is said to be in involution, or to form a system in involution with the equations or loci $U = 0$, $V = 0$. If U, V are binary quantics then the equation $U + kV = 0$ may be considered as representing a range of points in involution with the ranges $U = 0$, $V = 0$. And similarly, if U, V are ternary quantics, then the equation $U + kV = 0$ may be considered as representing a curve in involution with the curves $U = 0$, $V = 0$.

In the case of a range $U + kV = 0$, the constant k may be determined so that the range shall have a twofold[2] point. The condition for this may be written

$$\text{Disct.}\,.\,(U + kV) = 0,$$

it being understood that the discriminant of the function $U + kV$ is taken in regard to the coordinates. And this being so, we may write Disct.. Disct.. $(U+kV)$ to denote the discriminant in regard to k of the function Disct.. $(U+kV)$. The quantity in question (viz. Disct.. Disct.. $(U+kV)$, or say for shortness D) is a function of the coefficients of U, V, homogeneous as regards each set of coefficients separately, and it breaks up into factors in the form

$$D = RQ^2P^2;$$

where $R = 0$ is the condition in order that the ranges $U = 0$, $V = 0$ may have a point in common; or what is the same thing, in order that there may have a range $U + kV = 0$ having a twofold point at a common point of the ranges $U = 0$, $V = 0$, (R is in fact the resultant of the quantics U, V): $Q = 0$ is the condition in order that there may be a range $U + kV = 0$ having a threefold point: and $P = 0$ is the condition in order that there may be a range $U + kV = 0$ having a pair of twofold points.

[2]The series of epithets *onefold*, *twofold*, &c., seems preferable to the series *single*, *double*, &c., as avoiding ambiguities which would sometimes be occasioned by the use of these last. The double point of a curve I call a *node*, viz. a *crunode* when it is a double point with two real branches, and an *acnode* when it is a conjugate or isolated point. The subject-matter and context will in general show whether the term node is to be considered as including or as not including a cusp.

And similarly, when $U = 0$, $V = 0$ are curves, then we have the like equation

$$D = RQ^2 P^2,$$

where $R = 0$ is the condition in order that the curves $U = 0$, $V = 0$ may have a point of twofold intersection, that is, that the two curves may touch each other, (R is the Tactinvariant of the quantics U, V); or what is the same thing, it is the condition in order that there may be a curve $U + kV = 0$ having a node at a point of twofold intersection of the curves $U = 0$, $V = 0$; moreover $Q = 0$ is the condition in order that there may be a curve $U + kV = 0$ having a cusp: and $P = 0$ is the condition in order that there may be a curve $U + kV = 0$ having a pair of nodes.

The establishment and illustration of the foregoing theorems form the chief object of the present memoir.

Article Nos. 1 to 16, relating to two Binary Quantics.

1. Let $U = (a, \dots \mid x, y)^n$, $V = (a', \dots \mid x, y)^n$, be two binary quantics of the same order n, and write $W = U + kV = (a + ka', \dots \mid x, y)^n$, so that $W = U + kV = 0$ is the equation of a range in involution with the ranges $U = 0$, $V = 0$. But for greater distinctness it is in general convenient to retain $U + kV$ instead of replacing it by the single letter W.

2. In order that the range $U+kV = 0$ may have a twofold point we must have simultaneously

$$\partial_x(U + kV) = 0, \qquad \partial_y(U + kV) = 0,$$

and eliminating (x, y) from these equations we find

$$\text{Disct.}\,.\,(U + kV) = 0,$$

which is an equation of the order $2(n - 1)$ as regards k, and of the same order as regards the coefficients of U and V conjointly. And to each of the $2(n - 1)$ values of k there corresponds a point (x, y) satisfying the required conditions; that is, a point which is a twofold point of the range $U + kV = 0$. The points in question may be termed the 'critic centres' of the involution.

3. The elimination of k from the before-mentioned two equations gives

$$\begin{vmatrix} \partial_x U & \partial_y U \\ \partial_x V & \partial_y V \end{vmatrix} = 0,$$

where the determinant, which for shortness I call J, is the Jacobian of the two functions U, V. The equation $J = 0$ gives a range of $2(n - 1)$ points which are in fact the critic centres; and for each of these points we have

$$\partial_x U : \partial_x V = \partial_y U : \partial_y V = -k : 1,$$

which gives the value of k corresponding to the point in question.

4. The condition in order that the equation in k may have a twofold root is

$$\text{Disct.}\,.\,\text{Disct.}\,.\,(U + kV) = 0, \quad \text{or say} \quad D = 0,$$

where $\text{Disct.}\,.\,\text{Disct.}\,.\,(U + kV) = D$ is a function of the degree $2(2n - 2)(2n - 3)$ in regard to the coefficients of U, V conjointly; but it is separately homogeneous, and therefore of the degree $(2n - 2)(2n - 3)$ in regard to each of the two sets of coefficients.

5. To each point of the range $J = 0$ there corresponds a value of k; hence if the range J have a twofold point, then the equation in k will have a twofold root. Now first if the ranges $U = 0$,

$V = 0$ have a common point, then this is a twofold point of the range $J = 0$. But secondly, without a common point in the ranges $U = 0$, $V = 0$, the range $J = 0$ may have a twofold point; and in this case also without a twofold point in the range $J = 0$, there may be in this range two onefold points giving each of them the same value of k, and so giving a twofold root of the equation in k. And the three suppositions correspond respectively to the cases of there being a range $U + kV = 0$ having a twofold point at a common point of the ranges $U = 0$, $V = 0$, having a threefold point, and having a pair of twofold points.

6. First, if the ranges $U = 0$, $V = 0$ have a common point we may write

$$U = (x - \alpha y)U', \qquad V = (x - \alpha y)V',$$

and these give

$$U + kV = (x - \alpha y)(U' + kV').$$

Now in general

$$\text{Disct.}\,.\,PQ \;=\; \text{Disct.}\,.\,P \cdot \text{Disct.}\,.\,Q \cdot [\text{Result.}\,.(P,Q)]^2,$$

and hence in the present case

$$\text{Disct.}\,.\,(U + kV) \;=\; \text{Disct.}\,.\,(U' + kV') \cdot (U'_0 + kV'_0)^2,$$

where $U'_0 + kV'_0$ is what $U' + kV'$ becomes on writing therein $x = \alpha y$ and neglecting the factor y^{n-1} which then presents itself. We see therefore that in this case the equation k has a twofold root $k = -U'_0 : V'_0$; a result which might also have been obtained from the consideration of the Jacobian.

7. The condition in order that the ranges $U = 0$, $V = 0$ may have a common point is

$$\text{Result.}\,.\,(U, V) = 0,$$

say $R = 0$. R is of the degree n in regard to the coefficients of U, V respectively.

8. Secondly, suppose that the functions U, V are such that there exists a range $U + kV = 0$ having a threefold point. If k_1 be the proper value of k, then we have $U + k_1V = (x - \alpha y)^3\Theta$, and therefore $U = -k_1V + (x - \alpha y)^3\Theta$. Hence forming the Jacobian of U, V, the equation for the determination of the critic centres will be

$$\begin{vmatrix} \partial_x V & \partial_x (x - \alpha y)^3\Theta \\ \partial_y V & \partial_y (x - \alpha y)^3\Theta \end{vmatrix} = 0,$$

which is of the form $(x - \alpha y)^2\Omega = 0$; or we have $(x - \alpha y)^2 = 0$, a twofold critic centre. The corresponding value of k given by the equation $-k : 1 = \partial_x U : \partial_x V$ is $k = k_1$, and we have thus $k = k_1$ as a twofold root of the equation in k.

9. But if the range $W = U + kV = 0$ has a threefold point, or what is the same thing, if the equation $W = 0$ has a threefold root; then we must have between the coefficients of W a plexus of equations equivalent to two relations. Such plexus is known to be of the order $3(n - 2)$. This comes to saying that if the coefficients of W are assumed to be of the form $a + ka' + la'', \ldots$ and if between the several equations of the plexus we eliminate k, we obtain for l an equation $Q = 0$ of the degree $3(n-2)$. The equation in question would be of the form Funct.$.(a + la'', a', \ldots) = 0$. Hence Q is of the degree $3(n - 2)$ in the coefficients $(a, \ldots)$ of U. And in a similar manner Q is of the degree $3(n - 2)$ in the coefficients $(a', \ldots)$ of V. And omitting altogether the terms in l, or taking the coefficients of W to be $a + ka', \ldots$ if from the equations of the plexus we eliminate k, we find an equation $Q = 0$, where Q is a function of the degree $3(n - 2)$ as regards the coefficients of U, and of the same degree as regards the coefficients of V. We have thus found the form of the condition $Q = 0$ which expresses that there may be a range $U + kV = 0$ having a threefold point.

10. It may be proper to remark conversely that given the equation $Q = 0$, if in this equation we write $a = (a + ka') - ka', \dots$ so that Q becomes a function of $a + ka', \dots, a', \dots, k$, then the equation $Q = 0$ will be satisfied irrespectively of the values of $a, \dots, k$ by a plexus of equations involving only the coefficients $a + ka', \dots$ and which is in fact the very plexus (equivalent therefore to two relations) which gives the conditions in order that the equation $W = 0$ may have a threefold root.

11. Thirdly, suppose that the functions U, V are such that there exists a range $U + kV = 0$ having a pair of twofold points. If k_1 be the proper value of k, then we have $U + k_1V = (x - \alpha y)^2(x - \beta y)^2\Theta$, and therefore $U = -k_1V + (x - \alpha y)^2(x - \beta y)^2\Theta$. Hence forming the Jacobian of U, V, we have for the determination of the critic centres the equation

$$\begin{vmatrix} \partial_x V & \partial_x (x - \alpha y)^2(x - \beta y)^2\Theta \\ \partial_y V & \partial_y (x - \alpha y)^2(x - \beta y)^2\Theta \end{vmatrix} = 0,$$

which is of the form $(x - \alpha y)(x - \beta y)\Omega = 0$; or, we have $x - \alpha y = 0$, or $x - \beta y = 0$, a pair of critic centres; and for each of these the corresponding value of k given by the equation $-k : 1 = \partial_x U : \partial_x V$ is $k = k_1$, so that $k = k_1$ is a twofold root of the equation in k.

12. By the like considerations as for the threefold root (observing that if the equation $W = 0$ has a pair of twofold roots we must have between the coefficients of W a plexus equivalent to two relations, and of the order $2(n - 2)(n - 3)$), we see that the condition for the existence of a range $U + kV = 0$ having a pair of twofold points is of the form $P = 0$, where P is a function of the degree $2(n - 2)(n - 3)$ as regards the coefficients of U, and of the same degree as regards the coefficients of V; and conversely that, given the equation $P = 0$, we may find the plexus.

13. The equation $D = 0$ will be satisfied if $R = 0$, or if $Q = 0$, or if $P = 0$; and in no other cases. To prove this, suppose that $x - \alpha y = 0$ is the critic centre corresponding to a twofold root k_1 of the equation in k. We have $U = -k_1V + (x - \alpha y)^2\Theta$, and thence the equation for the critic centres is

$$\begin{vmatrix} \partial_x V & \partial_x (x - \alpha y)^2\Theta \\ \partial_y V & \partial_y (x - \alpha y)^2\Theta \end{vmatrix} = 0,$$

which is an equation of the form $(x - \alpha y)\Omega = 0$; and where, corresponding to the root $x - \alpha y = 0$, the equation $-k : 1 = \partial_x U : \partial_x V$ gives $k = k_1$. Since k_1 is a twofold root, there must be another critic centre also giving the value k_1 of k. This new critic centre may be either $x - \alpha y = 0$ (the same as the first mentioned critic centre) or it may be a distinct critic centre $x - \beta y = 0$. In the former case

$$\begin{vmatrix} \partial_x V & \partial_x (x - \alpha y)^2\Theta \\ \partial_y V & \partial_y (x - \alpha y)^2\Theta \end{vmatrix}$$

must contain, instead of the factor $(x - \alpha y)$, the factor $(x - \alpha y)^2$. In order that this may be so, we must have $(\alpha\partial_x V + \partial_y V)\Theta$ divisible by $(\partial_x V \cdot \alpha\partial_y V)$, that is, either $\alpha\partial_x V + \partial_y V$, or else Θ, divisible by $x - \alpha y$; or, what is the same thing, either V, or else Θ, divisible by $x - \alpha y$. But if V be divisible by $x - \alpha y$, then U, V have the common factor $x - \alpha y$, and we have the case first above considered. And again if Θ contain the factor $x - \alpha y$, then we have

$$U = -k_1V + (x - \alpha y)^3\Theta',$$

and we have the case secondly above considered. Finally if the new critic centre be the distinct centre $x - \beta y = 0$, then for $x - \beta y = 0$ the equation

$$-k : 1 = \partial_x U : \partial_x V = \partial_y U : \partial_y V$$

should give $k = k_1$; but this will only happen if $\partial_x \cdot (x-\alpha y)^2\Theta$, $\partial_y \cdot (x-\alpha y)^2\Theta$ vanish for $x-\beta y = 0$, that is, if Θ contains the factor $(x-\beta y)^2$; and when this is so,

$$U = -k_1 V + (x-\alpha y)^2(x-\beta y)^2\Theta',$$

or we have the case thirdly above considered.

14. Hence the equation $D = 0$ being satisfied if $R = 0$, or else if $Q = 0$, or else if $P = 0$, and in no other cases, the function D must be made up of the factors R, Q, P, each taken the proper number of times, and knowing the degrees of the several functions, it follows that we must have

$$D \;=\; RQ^2P^2;$$

in fact, considering the coefficients of either U or V, the comparison of the degrees gives

$$2(n-1)(2n-3) \;=\; n + 9(n-2) + 4(n-2)(n-3),$$

where the function on the right-hand side is

$$n + 9n - 18 + 4n^2 - 20n + 24 \;=\; 4n^2 - 10n + 6,$$

which is the value of the function on the left-hand side.

15. In the very particular case $n = 2$, Q and P are each of them of the degree $= 0$; and we have simply $D = R$, that is, the resultant of the two quadric functions

$$U = (a, b, c \mid x, y)^2, \qquad V = (a', b', c' \mid x, y)^2,$$

is

$$= \text{Disct.}.(U + kV) \;=\; \text{Disct.}.\big((ac - b^2,\;\; ac' + a'c - 2bb',\;\; a'c' - b'^2) \mid 1, k\big)^2,$$

which is Prof. Boole's ancient theorem referred to in my Fifth Memoir on Quantics[3], but which is now first exhibited in connexion with the general theory to which it belongs.

16. It may be noticed that the condition for a twofold critic centre, or (what is the same thing) a twofold factor of the Jacobian (which condition is of the degree $2(2n-3)$ in regard to the coefficients of U or V) is $RQ = 0$; and that we in fact have

$$2(2n-3) \;=\; n + 3(n-2).$$

This remark is due to Dr Salmon.

Article Nos. 17 to 42, relating to two Ternary Quantics.

17. Suppose now that $U = (a, \dots \mid x, y, z)^n$, $V = (a', \dots \mid x, y, z)^n$ are two ternary quantics of the same order n, and write $W = U + kV = (a + ka', \dots \mid x, y, z)^n$, so that

$$W = U + kV = 0$$

is the equation of a curve in involution with the curves $U = 0$, $V = 0$. But for greater distinctness it is in general proper to retain $U + kV$ in place of W.

18. In order that the curve $U + kV = 0$ may have a node, we must have simultaneously

$$\partial_x(U + kV) = 0, \qquad \partial_y(U + kV) = 0, \qquad \partial_z(U + kV) = 0,$$

[3] *Phil. Trans.* vol. CXLVIII. (1858), pp. 415–427, [156].

and eliminating (x, y, z) from these equations we have

$$\text{Disct.}\,.\,(U + kV) \ = \ 0,$$

which is an equation of the degree $3(n-1)^2$ as regards k, and of the same order as regards the coefficients of U, V conjointly.

19. To each of the $3(n-1)^2$ values of k there corresponds a point satisfying the conditions in question, and which is therefore a node of the corresponding nodal curve $U + kV = 0$; the points in question are the critic centres of the involution.

20. The critic centres may be differently obtained as follows; viz. if from the three equations we eliminate k, we find

$$\begin{vmatrix} \partial_x U & \partial_y U & \partial_z U \\ \partial_x V & \partial_y V & \partial_z V \end{vmatrix} = 0,$$

a plexus of three curves, each of them of the order $2(n-1)$; any two of the three curves intersect in $4(n-1)^2$ points; but $(n-1)^2$ of these do not lie on the third curve; the remaining $3(n-1)^2$ of them lie on all three of the curves, and they are the critic centres of the involution.

21. More generally the critic centres lie on any curve whatever of the form

$$\begin{vmatrix} \alpha & \beta & \gamma \\ \partial_x U & \partial_y U & \partial_z U \\ \partial_x V & \partial_y V & \partial_z V \end{vmatrix} = 0,$$

and any such curve, viz. any curve of the order $2(n-1)$ passing through the $3(n-1)^2$ critic centres, may be termed a *diacritic* curve.

22. For any one of the critic centres we have

$$\partial_x U : \partial_y U : \partial_z U = \partial_x V : \partial_y V : \partial_z V = -k : 1,$$

which gives the value of k corresponding to the point in question.

23. The condition in order that the equation in k may have a twofold root is

$$\text{Disct.}\,.\ \text{Disct.}\,.\,(U + kV) = 0, \quad \text{or say} \quad D = 0,$$

where $\text{Disct.}\,.\ \text{Disct.}\,.\,(U + kV) = D$ is a function of the degree $2 \cdot 3(n-1)^2\{3(n-1)^2 - 1\}$ in regard to the coefficients of U, V conjointly; but it is separately homogeneous, and therefore of the degree $3(n-1)^2\{3(n-1)^2 - 1\}$ in regard to each set of coefficients.

24. To each of the critic centres there corresponds a value of k. Hence if two of the critic centres coincide, or say if there is a twofold critic centre, the equation in k will have a twofold root. Now first if the curves $U = 0$, $V = 0$ touch each other (have a point of contact or twofold intersection), then the diacritic curves will all touch (have a point of twofold intersection) at the point in question, which is therefore a twofold critic centre. It may be remarked in passing that the diacritic curves do not at the twofold critic centre touch the curves $U = 0$, $V = 0$. But secondly, the diacritic curves may touch at a point which is not a point of contact of the curves $U = 0$, $V = 0$. Such a point is a twofold critic centre. In each of these two cases the equation in k has a twofold root. Moreover, in the first case the curve $U + kV = 0$ corresponding to the twofold root has a node at the point of contact of the two curves $U = 0$, $V = 0$; in the second case the curve $U + kV = 0$ corresponding to the twofold root has the twofold centre (not a mere node but) a cusp. And thirdly, without any twofold critic centre, two distinct critic centres may give by the equations

$$\partial_x U : \partial_y U : \partial_z U = \partial_x V : \partial_y V : \partial_z V = -k : 1$$

the same value of k, and then the curve $U + kV = 0$ corresponding to such value of k is a curve having a node at each of the critic centres in question, that is, it has two nodes.

25. First, if the curves $U = 0$, $V = 0$ touch each other, then, (x, y, z) being the coordinates of the point of contact, we have

$$\partial_x(U + k_1V) = 0, \quad \partial_y(U + k_1V) = 0, \quad \partial_z(U + k_1V) = 0,$$

where k_1 denotes the value given by the equations

$$\partial_xU : \partial_yU : \partial_zU = \partial_xV : \partial_yV : \partial_zV = -k_1 : 1$$

belonging to the point of contact. It at once follows that every diacritic curve passes through the point in question. But it is somewhat more difficult to show that the diacritic curves touch at this point.

26. I represent for shortness the first and second differential coefficients of U by (L, M, N), (a, b, c, f, g, h), and similarly those of V by (L', M', N'), (a', b', c', f', g', h'), these values all belonging to the point of contact: we have therefore

$$L + k_1L' = 0, \quad M + k_1M' = 0, \quad N + k_1N' = 0.$$

The equation of the diacritic curve is

$$\begin{vmatrix} \alpha & \beta & \gamma \\ L & M & N \\ L' & M' & N' \end{vmatrix} = 0;$$

to find the tangent we must operate on the left-hand side with $X\partial_x + Y\partial_y + Z\partial_z$, where X, Y, Z are current coordinates. Calling the foregoing symbol D, this gives, after development and substitution *[see facsimile pp. 303–304 for full computations]*, the tangent equation

$$\begin{vmatrix} \alpha & \beta & \gamma \\ L + \theta L' & M + \theta M' & N + \theta N' \\ DL + k_1DL' & DM + k_1DM' & DN + k_1DN' \end{vmatrix} = 0,$$

or, substituting for D its value, the equation of the tangent is

$$\begin{vmatrix} \alpha & \beta & \gamma \\ L + \theta L' & M + \theta M' & N + \theta N' \\ (a + ka', \ldots \mid X, Y, Z) & (h + kh', \ldots \mid X, Y, Z) & (g + kg', \ldots \mid X, Y, Z) \end{vmatrix} = 0.$$

27. Now if the diacritics touch, this equation should be independent of α, β, γ. Putting for shortness $a + k_1a' = a$, &c., and also taking as we may do $\theta = 0$, the parts multiplied by α, β, γ respectively are

$$M(gX + fY + cZ) - N(hX + bY + fZ),$$
$$N(aX + hY + gZ) - L(gX + fY + cZ),$$
$$L(hX + bY + fZ) - M(aX + hY + gZ),$$

and we ought therefore to have

$$Mg - Nh : Mf - Nb : Mc - Nf = Na - Lg : Nh - Lf : Ng - Lc = Lh - Ma : Lb - Mh : Lf - Mg.$$

These equations are in fact satisfied; for take any one of them, for example, the equation

$$\frac{Mg - Nh}{Na - Lg} = \frac{Mf - Nb}{Nh - Lf};$$

this is

$$MN(gh - af) - N^2(h^2 - ab) - LM(fg - fg) + LN(fh - bg) \;=\; 0,$$

or, omitting the term in LM, and throwing out the factor N,

$$L(hf - bg) + M(gh - af) + N(ab - h^2) \;=\; 0.$$

But the equations $L + kL' = 0$, &c., give

$$ax + hy + gz = 0, \qquad hx + by + fz = 0, \qquad gx + fy + cz = 0,$$

that is

$$x : y : z \;=\; bc - f^2 : fg - ch : hf - bg \;=\; fg - ch : ca - g^2 : gh - af \;=\; hf - bg : gh - af : ab - h^2,$$

so that the above written equation is $Lx + My + Nz = 0$, which is true in virtue of the equation $U = 0$; and similarly for all the other equations which were to be verified.

28. It is to be noticed that the determination of the tangent of the diacritics depends only on the second differential coefficients (a, b, c, f, g, h), (a', b', c', f', g', h') of U, V. The tangent in question will be the same if instead of the curves $U = 0$, $V = 0$ we have the conics passing through the point of contact of the two curves, and whose tangents are coincident with those of the two curves $U = 0$, $V = 0$ respectively; that is, the conics touch at the point in question. They consequently intersect in two more points; the chord of intersection or line joining the last-mentioned two points meets the common tangent in a point; the polars of this point in regard to the two conics respectively pass through the point of contact, and moreover they are one and the same line; this line is the required tangent of the diacritics. The proof will be given, *post*, No. 41.

29. Let $R = 0$ be the condition in order that the two curves $U = 0$, $V = 0$ may touch each other, or say let R be the Tactinvariant of U, V. When the curves U, V are of the degrees m, n respectively, then R is of the degrees $n(2m + n - 3)$, $m(m + 2n - 3)$ in regard to the coefficients of U, V respectively. Hence in the present case where U, V are each of the degree n, R is of the degree $3n(n - 1)$ in regard to each set of coefficients.

30. Secondly, if the functions U, V are such that there exists a curve $U + kV = 0$ (say the curve $U + k_1V = 0$) which has a cusp, then it is to be shown that $k = k_1$ is a twofold root of the equation in k; and to do this it has to be shown that the cusp is a twofold critic centre; or that the diacritic curves touch at the cusp: it may be added that the cuspidal tangent is the common tangent of the diacritic curves. Now the cuspidal curve being $U + k_1V = 0$, then at the cusp the first derived functions $L + k_1L'$, $M + k_1M'$, $N + k_1N'$ vanish identically; and moreover the second derived functions $a + k_1a'$, ... are such that (X, Y, Z) being any magnitudes whatever, $(a + k_1a', \ldots \mid X, Y, Z)^2$ is a perfect square, $= (\lambda X + \mu Y + \nu Z)^2$ suppose. Now X, Y, Z being current coordinates, and D denoting the operation $D = X\partial_x + Y\partial_y + Z\partial_z$, the equation of the tangent to the diacritic curve (by an investigation similar to that for this same tangent in the case first above considered) is found to be

$$\begin{vmatrix} \alpha & \beta & \gamma \\ L + \theta L' & M + \theta M' & N + \theta N' \\ (a + k_1a', \ldots \mid X, Y, Z) & (h + k_1h', \ldots \mid X, Y, Z) & (g + k_1g', \ldots \mid X, Y, Z) \end{vmatrix} = 0,$$

and we have

$$(a + k_1a', \ldots \mid X, Y, Z) = \tfrac{1}{2}\partial_X(a + k_1a', \ldots \mid X, Y, Z)^2 = \tfrac{1}{2}\partial_X(\lambda X + \mu Y + \nu Z)^2 = \lambda(\lambda X + \mu Y + \nu Z),$$

and similarly the values of $(h + k_1h', \dots \mid X, Y, Z)$ and $(g + k_1g', \dots \mid X, Y, Z)$ are $= \mu(\lambda X + \mu Y + \nu Z)$ and $= \nu(\lambda X + \mu Y + \nu Z)$ respectively. Hence the equation of the tangent to the diacritic curve is

$$\lambda X + \mu Y + \nu Z = 0,$$

that is, the tangent being independent of the values of (α, β, γ) is the same for all the diacritic curves, and is the tangent at the cusp of the cuspidal curve $U + k_1V = 0$.

31. The conditions in order that the curve $W = U + kV = 0$ may have a cusp are given by a plexus equivalent to three relations between the coefficients $a + ka', \dots$ of W, and using for a moment β to denote the degree of the plexus or system, then eliminating k between the equations of the plexus we find between the coefficients $a, \dots$ of U and the coefficients $a', \dots$ of V an equation $Q = 0$ of the degree β in regard to the two sets of coefficients respectively. Conversely, given the equation $Q = 0$, we may find the plexus between the coefficients $a+ka', \dots$ of W. The value of β, as will be shown *post*, Annex, is

$$= 12(n-1)(n-2).$$

32. Thirdly, when the functions U, V are such that there exists a curve $U+kV = 0$ (suppose the curve $U + k_1V = 0$) which has a pair of nodes; each of these nodes is a critic centre, and (by means of the equation $-k : 1 = \partial_xU : \partial_xV$) gives the value k_1 of k, that is, k_1 is a twofold root of the equation in k.

33. The conditions in order that the curve $W = U + kV = 0$ may have a pair of nodes are given by a plexus of the degree α; then the coefficients being $a + ka', \dots$ if we eliminate k between the equations of the plexus, we find between the coefficients $a, \dots$ of U and $a', \dots$ of V an equation $P = 0$ of the degree α in the two sets of coefficients respectively. And conversely, given the equation $P = 0$, we may find the plexus between the coefficients $a + ka', \dots$ of W. I have not succeeded in finding directly the value of α, but only derive it from the equation $D = RQ^2P^2$, which if α had been found independently, would have been verified by means of such value of α; the value is

$$\alpha \;=\; \tfrac{1}{2} \cdot 3(n-1)(n-2)(3n^2 - 3n - 11).$$

34. The equation $D = 0$ is satisfied if $R = 0$, or if $Q = 0$, or if $P = 0$, and it may be seen that it is not satisfied in any other case. Hence D is made up of the factors P, Q, R, and I assume that its form is the same as in the case of a binary quantic, that is, that we have

$$D \;=\; RQ^2P^2.$$

35. Comparing the degrees of the two sides we have

$$3(n-1)^2(3n^2 - 6n + 2) \;=\; 3n(n-1) + 36(n-1)(n-2) + 2(n-1)(n-2)(3n^2 - 3n - 11);$$

or, what is the same thing,

$$(n-1)(3n^2 - 6n + 2) \;=\; n + 12(n-2) + (n-2)(3n^2 - 3n - 11),$$

which is true, but, as just remarked, this equation itself was used to find the value

$$\alpha \;=\; \tfrac{1}{2} \cdot 3(n-1)(n-2)(3n^2 - 3n - 11).$$

36. Recapitulating, the equation in k will have a twofold root

1°. if $R = 0$, that is, if the curves $U = 0$, $V = 0$ touch each other, and in this case there is a twofold critic centre at the point of contact;

2°. if $Q = 0$, that is, if there be a curve $U + kV = 0$ having a cusp, and in this case the cusp is a twofold critic centre;

3°. if $P = 0$, that is, if there is a curve $U + kV = 0$ having a pair of nodes.

37. The three cases may be geometrically illustrated by supposing that the curves $U = 0$, $V = 0$ are in the first instance nearly, but not exactly, in the several relations in question.

First, if the curves $U = 0$, $V = 0$ are about to touch each other, that is, if there are two points of intersection about to coincide with each other. There are here two critic centres in the neighbourhood of the two points of intersection, and which, when the two points of intersection become a point of contact, coincide each with the point of contact.

Secondly, when the curves are such that there are two critic centres which become ultimately a twofold centre.

Thirdly, when the curves are such that there are two critic centres which remaining distinct from each other belong ultimately to the same critic curve.

38. The curves 1 and 2 in the left-hand figures respectively represent the nodal curves corresponding to slightly different values of k, which in the right-hand figures respectively give the curve corresponding to a twofold value of k. In the first pair of figures the curves $U = 0$ and $V = 0$, about to touch in the left-hand figure, touching in the right-hand figure, are shown by dotted lines. It will be observed that in the second case in the left-hand figure the two nodes which give rise to a cusp are the one of them an acnode and the other a crunode; this is in fact the only mode of drawing the figure so that a cusp shall present itself. The transition of form is one of ordinary occurrence in cubic curves and in curves of a higher order; thus if $y^2 = (x - a)(x - b)(x - c)$, where $a < b < c$, then if $a = b$, we have an acnodal curve, if $b = c$ a crunodal curve, and if $a = b = c$ a cuspidal curve.

39. In the case of two conics, $n = 2$. We have here simply $D = R$, where $R = 0$ is the condition in order that the two conics may touch each other. The nodal curves are of course the three pairs of lines passing through the points of intersection of the two conics, and the nodes of these curves, or critic centres, are the centres of the quadrangle formed by the four points in question; or, what is the same thing, they are the system of conjugate points common to the two conics, viz. the points which are such that the polar of one of them taken with respect to either of the conics is the line joining the other two of them. The diacritics are any conics passing through the three points.

40. If the two conics are

$$(a, b, c, f, g, h \mid x, y, z)^2 = 0, \qquad (a', b', c', f', g', h' \mid x, y, z)^2 = 0,$$

and if the determinant formed with $a + ka'$, &c., is denoted by $(K, \Theta, \Theta', K' \mid 1, k)^3$, so that

$$K = abc - af^2 - bg^2 - ch^2 + 2fgh,$$

$$\Theta = a'(bc - f^2) + \text{\&c.},$$

$$\Theta' = a(b'c' - f'^2) + \text{\&c.},$$

$$K' = a'b'c' - a'f'^2 - b'g'^2 - c'h'^2 + 2f'g'h',$$

then the before-mentioned equation $D = R = 0$, which gives the condition that the conics may touch, is

$$\text{Disct.}\,.\,(K, \Theta, \Theta', K' \mid 1, k)^3 \;=\; 0,$$

where the left-hand side is of the order 6 in the coefficients of the two conics respectively: this is a known formula.

41. If the equation in k have a twofold root, the two conics will touch: two of the critic centres will then coincide at the point of contact, or this point is a twofold critic centre: the remaining or onefold critic centre is the intersection of the common tangent and of the line joining the two points of intersection of the conics. In virtue of the general property, the first-mentioned two centres must be considered as lying on the line which is the polar of the onefold critic centre in regard to either of the conics. The diacritics pass through the critic centres, that is, they pass through the onefold centre, and touch the polar in question at the point of contact of the two conics, or twofold critic centre; this is in fact the property mentioned *ante*, No. 28.

42. The equation

$$\text{Disct.. Disct..}(U + kV) \;=\; 0,$$

as applied to a conic and a circle leads at once to the equation of the curve parallel to a given conic; such parallel curve is in fact the envelope of the circles of a given constant radius which touch the given conic. This method is in effect due to Dr Salmon, who applied the corresponding theorem *in solido* to the determination of the surface parallel to an ellipsoid.

Annex, referred to No. 31. Investigation of the order of the plexus or system for the existence of a Cusp.

Considering for a moment the curve

$$U = (* \mid x, y, z)^n = 0, \quad \text{let } (L, M, N),\ (a, b, c, f, g, h)$$

be the first and second differential coefficients of U; (A, B, C, F, G, H) the inverse system, viz. $A = bc - f^2$, &c. At a cusp we have

$$L = 0,\ M = 0,\ N = 0,\ A = 0,\ B = 0,\ C = 0,\ F = 0,\ G = 0,\ H = 0,$$

a system of equations which is contained in the system $L = 0$, $M = 0$, $N = 0$, $A = 0$. But this system contains besides the cusp system the irrelevant system $L = 0$, $M = 0$, $N = 0$, $x^2 = 0$. In fact the equations $L = 0$, $M = 0$, $N = 0$ give

$$ax + hy + gz = 0, \quad hx + by + fz = 0, \quad gx + fy + cz = 0,$$

and thence

$$x^2 : y^2 : z^2 : yz : zx : xy = A : B : C : F : G : H.$$

Hence the equation $A = 0$, if $x^2 = 0$, implies the entire system

$$A = 0,\ B = 0,\ C = 0,\ F = 0,\ G = 0,\ H = 0.$$

But if $L = 0$, $M = 0$, $N = 0$, $x^2 = 0$, then these equations give $A = 0$ (and also $H = 0$, $G = 0$), but they do not give the remaining equations $B = 0$, $C = 0$, $F = 0$. Or the same thing may be shown in a less symmetrical form, but more clearly thus; we have identically

$$-cx(ax + hy + gz) + (hx - fz)(hx + by + fz) + hz(gx + fy + cz) - (bc - f^2)z^2 + (ab - h^2)x^2 = 0,$$

whence the equations $L = 0$, $M = 0$, $N = 0$, $A = 0$ give $Cx^2 = 0$, that is, $C = 0$, or $x^2 = 0$. But the equations $L = 0$, $M = 0$, $N = 0$, $A = 0$, $C = 0$ give (as it is easy to show) the entire system $A = 0$, $B = 0$, $C = 0$, $F = 0$, $G = 0$, $H = 0$. That is, the system

$$L = 0,\ M = 0,\ N = 0,\ A = 0$$

is made up of the cusp system, and of the system ($L = 0,\ M = 0,\ N = 0,\ A = 0,\ x^2 = 0$); or since $A = 0$ is a consequence of the other equations, the second system is

$$(L = 0,\ M = 0,\ N = 0,\ x^2 = 0).$$

Consider now the curve $\lambda U + \mu U' + \nu U'' = 0$, which will have a cusp if the ratios $\lambda : \mu : \nu$ are properly determined. And to each set of values of $\lambda : \mu : \nu$ there corresponds a set of values (x, y, z), the coordinates of a cusp of the curve; so that the number of such sets, that is, the number of points each whereof is the cusp of a corresponding curve $\lambda U + \mu V + \nu W = 0$ is precisely equal to the number of sets of values of $\lambda : \mu : \nu$; or it is equal to the order of the system of conditions for the existence of a cusp.

Denoting as before the first and second differential coefficients of U by $L, M, N, a, b, c, f, g, h$, and those of U', U'' in a corresponding manner, and taking for the cusp the system before represented by $L = 0$, $M = 0$, $N = 0$, $A = 0$, we have

$$\lambda L + \mu L' + \nu L'' = 0,$$

$$\lambda M + \mu M' + \nu M'' = 0,$$

$$\lambda N + \mu N' + \nu N'' = 0,$$

$$(\lambda b + \mu b' + \nu b'')(\lambda c + \mu c' + \nu c'') - (\lambda f + \mu f' + \nu f'')^2 = 0,$$

which last equation, to denote that it is of the second order in regard to the differential coefficients a, b, &c., a', &c., I represent by

$$((a, b, \ldots) \mid \lambda, \mu, \nu)^2 = 0.$$

But this system of four equations contains not only the cusp system, but the system made of the three linear equations and the equation $x^2 = 0$. Eliminating λ, μ, ν, the last-mentioned system is

$$\begin{vmatrix} L & L' & L'' \\ M & M' & M'' \\ N & N' & N'' \end{vmatrix} = 0, \qquad x^2 = 0,$$

where the first equation is that of a curve of the order $3(n-1)$. And the two equations give together $6(n-1)$ points, viz. the points of intersection of the curve by the line $x = 0$, each reckoned as a twofold point.

Returning to the first-mentioned system, this may be replaced by

$$\begin{vmatrix} L & L' & L'' \\ M & M' & M'' \\ N & N' & N'' \end{vmatrix} = 0, \qquad ((a, b, \ldots) \mid L'M'' - L''M',\ \ L''M - LM'',\ \ LM' - L'M)^2 = 0,$$

which are curves of the orders $3(n-1)$ and $6n-8$ respectively. But each of these curves passes through the $3(n-1)^2$ points given by the equations

$$L = 0,\ M = 0,\ N = 0;$$

and these points are moreover nodes on the curve of the order $6n-8$; hence the points in question reckon as $6(n-1)^2$ intersections of the two curves. The number of the remaining intersections is

$$3(n-1)(6n-8) - 6(n-1)^2 \ = \ 6(n-1)\big(3n - 4 - (n-1)\big) \ = \ 6(n-1)(2n-3),$$

but among these are included the $6(n-1)$ intersections of the curve of the order $3(n-1)$ by the twofold line $x^2 = 0$; or, subtracting these, the number of the remaining points is

$$6(n-1)(2n-3-1) \ = \ 12(n-1)(n-2);$$

which number is consequently the order of the cusp system.

It may be remarked that considering the entire series of equations at first denoted by $(L = 0,\ M = 0,\ N = 0)$, $(A = 0,\ B = 0,\ C = 0,\ F = 0,\ G = 0,\ H = 0)$, the elimination of λ, μ, ν from the three linear equations gives as before

$$\begin{vmatrix} L & L' & L'' \\ M & M' & M'' \\ N & N' & N'' \end{vmatrix} = 0,$$

which is a curve of the order $3(n-1)$: and the eliminating of $\lambda^2, \mu^2, \nu^2,\ \mu\nu, \nu\lambda, \lambda\mu$ from the six quadric equations gives

$$\left| bc - f^2,\ \ b'c' - f'^2,\ \ b''c'' - f''^2,\ \ b'c'' + b''c' - 2f'f'',\ \ b''c + bc'' - 2ff'',\ \ bc' + b'c - 2ff' \right| = 0,$$

which is a curve of the order $12(n-2)$; the two curves would intersect in $36(n-1)(n-2)$ points, but as this is precisely three times the number $12(n-1)(n-2)$, I infer that these are in fact the $12(n-1)(n-2)$ points three times repeated, that is, that each of these is a point of threefold intersection of the two curves.

Cambridge, 7th November, 1863.

---

349. On a Case of the Involution of Cubic Curves *(begins)*

[From the Transactions of the Cambridge Philosophical Society, vol. XI. Part I. (1866), pp. 39–80.—Read 22 February, 1864.]

The present memoir relates to the involution

$$xyz + k(x+y+z)^2(\lambda x + \mu y + \nu z) \ = \ 0,$$

viz. treating x, y, z as coordinates, and k as a variable parameter, this equation represents the series of cubic curves passing through the intersections of the two cubics

$$xyz = 0, \qquad (x+y+z)^2(\lambda x + \mu y + \nu z) = 0;$$

or, what is the same thing, the line $x+y+z = 0$ meets any cubic of the series in three points the tangents at which are $x = 0$, $y = 0$, $z = 0$, and these tangents again meet the cubic in three points lying on the line $\lambda x + \mu y + \nu z = 0$; so that in the language which I have used elsewhere, the lines $x+y+z = 0$, $\lambda x + \mu y + \nu z = 0$ are in regard to the cubic a *primary* and a *satellite* line respectively. The investigation (which is a development of two short papers already published in the *Philosophical Magazine*)[4] was undertaken in order to applying it to the explanation and discussion of Plücker's Classification of Curves of the Third Order; but such application will properly be made in a separate memoir, *On the Classification of Cubic Curves*, and it has also

[4] *On the Cubic Centres of a Line with respect to Three Lines and a Line.—Phil. Mag.* t. XX. pp. 418–423 (1860), [257]. *Ditto, Second Paper*, t. XXII. pp. 433–436 (1861), [315].

appeared to me convenient to give therein the discussion of the geometrical forms of certain loci which present themselves in the present memoir.

I remark that the involution intended to be here considered is a case of the more general one $U + kV = 0$, where $U = 0$, $V = 0$ are any two cubic curves whatever. It appears from my memoir *On the Theory of Involution*, [348], that the equation Disct.. $(U + kV) = 0$, which determines the critic values of k is in the general case of the order 12; the special case is however in the present memoir treated irrespectively of the general one, and the equation for the critic values of k is found to be of the order 3; this of course means that the equation of the order 12 breaks up into two equations of the orders 9 and 3 respectively, but I have not attempted to show how the decomposition and reduction arise. Moreover, in the general case the equation Disct.. Disct.. $(U + kV) = 0$, which is the condition for the existence of a twofold critic value, presents itself in the form $RQ^2P^2 = 0$, where $R = 0$ is the condition that the two cubics $(U = 0, V = 0)$ shall touch each other; $Q = 0$ the condition that there shall be in the involution $U + kV = 0$ a curve having (not a mere node but) a cusp; and $P = 0$ the condition that there shall be a curve having two nodes, or (what is the same thing) breaking up into a line and conic. But in the special case, which, as already noticed, is here considered irrespectively of the general one, the equation Disct.. Disct.. $(U + kV) = 0$, for the existence of a twofold critic value presents itself in the reduced form $Q = 0$, giving the condition that corresponding to the twofold critic value there shall be a curve having (not a mere node but) a cusp.

Article Nos. 1 to 18, Explanations, Definitions, and Results.

1. I consider the involution

$$xyz + k(x + y + z)^2(\lambda x + \mu y + \nu z) \;=\; 0,$$

where $x = 0$, $y = 0$, $z = 0$, $x + y + z = 0$ may be considered as representing any four lines no three of which meet in a point, and $\lambda x + \mu y + \nu z = 0$ as representing any fifth line whatever: k is a variable parameter. The lines $x + y + z = 0$, $\lambda x + \mu y + \nu z = 0$, are a primary line and a satellite line of any cubic of the series, viz. the tangents $x = 0$, $y = 0$, $z = 0$ at the points of intersection with the primary line $x + y + z = 0$ meet the cubic in three points lying on the satellite line $\lambda x + \mu y + \nu z = 0$.

2. A *critic value* of k is a value for which the corresponding curve

$$xyz + k(x + y + z)^2(\lambda x + \mu y + \nu z) \;=\; 0$$

has a node; and such node, or say rather the site of such node, is a *critic centre.*

3. The critic values of k are in effect determined by a cubic equation, and the coordinates of the critic centre are then given rationally in terms of k; there are consequently three critic values of k; and the same number of critic centres, and of nodal curves: it is however found to be convenient to express as well the critic value of k, as the coordinates of the critic centre, rationally in terms of an auxiliary parameter θ, which is given by a cubic equation.

4. The cubic equation in k (or what is the same thing, that in θ) may have a twofold root (pair of equal roots); or, say rather, it may have a twofold root and a one-with-twofold root: corresponding to the twofold value of k we have a twofold critic centre, which is not a mere node but a cusp on the cubic, or instead of a merely nodal cubic we have a cuspidal cubic; and corresponding to the one-with-twofold value of k we have a one-with-twofold critic centre, being of course a mere node on the nodal cubic.

5. In the case in question of a twofold and one-with-twofold value of k, the line $\lambda x+\mu y+\nu z = 0$, or say the satellite line, envelopes a curve which might be termed the twofold and one-with-twofold envelope, but which is spoken of simply as the *envelope.*

The locus of the twofold centre is a curve which is called the *twofold centre locus.*

The locus of the one-with-twofold centre is a curve which is called the *one-with-twofold centre locus.*

These definitions premised, the following results may be stated:

6. The equation in θ may be represented in the three equivalent forms

$$\frac{1}{\theta+\lambda}+\frac{1}{\theta+\mu}+\frac{1}{\theta+\nu}-1=0, \qquad \frac{1}{\theta}=\frac{\theta}{\theta+\lambda}+\frac{\theta}{\theta+\mu}+\frac{\theta}{\theta+\nu},$$

$$\theta^3-\theta(\mu\nu+\nu\lambda+\lambda\mu)-2\lambda\mu\nu=0.$$

7. The critic value of k and the coordinates of the critic centre are then given by the equations

$$k=\frac{-\frac{1}{2}\theta^3}{(\theta+\lambda)(\theta+\mu)(\theta+\nu)},$$

$$x:y:z:x+y+z:\lambda x+\mu y+\nu z=\frac{1}{\theta+\lambda}:\frac{1}{\theta+\mu}:\frac{1}{\theta+\nu}:\frac{1}{\theta}:1.$$

8. The condition for a twofold and one-with-twofold value of k is

$$\sqrt{\lambda}+\sqrt{\mu}+\sqrt{\nu}=0,$$

or, what is the same thing,

$$(\mu\nu+\nu\lambda+\lambda\mu)^2-27\,\lambda^2\mu^2\nu^2=0,$$

which equations may either of them be considered as the line-equation of the envelope. The equation in the coordinates (x,y,z), or point-equation of the envelope is

$$\sqrt{x}+\sqrt{y}+\sqrt{z}=0,$$

or, in its rationalised form,

$$x^4+y^4+z^4-4(yz^3+y^3z+zx^3+z^3x+xy^3+x^3y)+6(y^2z^2+z^2x^2+x^2y^2)-124(x^2yz+xy^2z+xyz^2)=0.$$

9. The equation of the twofold centre locus is

$$\sqrt{x}+\sqrt{y}+\sqrt{z}=0,$$

or, in its rationalised form,

$$x^2+y^2+z^2-2yz-2zx-2xy=0;$$

the curve is therefore a conic, and it may be spoken of as the *twofold centre conic.*

10. The equation of the one-with-twofold centre locus is

$$x^3+y^3+z^3-(yz^2+y^2z+zx^2+z^2x+xy^2+x^2y)+3xyz=0;$$

the curve is therefore a cubic, and it may be spoken of as the *one-with-twofold centre cubic.*

11. The before-mentioned equation $\lambda^{-1/2}+\mu^{-1/2}+\nu^{-1/2}=0$ is satisfied by

$$\lambda:\mu:\nu=\alpha^{-2}:\beta^{-2}:\gamma^{-2},$$

where $\alpha+\beta+\gamma=0$, and it is very convenient to introduce these quantities α,β,γ into the formulae.

12. The equation of the satellite line giving a twofold and one-with-twofold centre is

$$\frac{x}{\alpha^2} + \frac{y}{\beta^2} + \frac{z}{\gamma^2} = 0,$$

the coordinates of the point of contact with the envelope are $x : y : z = \alpha^4 : \beta^4 : \gamma^4$.

The equation in θ gives $\theta_1 = \theta_2 = -\frac{1}{\alpha\beta\gamma}$ for the values corresponding to the twofold centre; and $\theta_3 = \frac{2}{\alpha\beta\gamma}$ for the value corresponding to the one-with-twofold centre.

The coordinates of the twofold centre, or cusp, are $x : y : z = \alpha^3 : \beta^3 : \gamma^3$.

The coordinates of the one-with-twofold centre, or node, are

$$x : y : z = \alpha^2(\beta - \gamma) : \beta^2(\gamma - \alpha) : \gamma^2(\alpha - \beta).$$

The equation of the tangent at the cusp is

$$(\beta - \gamma)\frac{x}{\alpha} + (\gamma - \alpha)\frac{y}{\beta} + (\alpha - \beta)\frac{z}{\gamma} = 0.$$

The equation of the line joining the cusp and the node, which line is also one of the tangents at the node, is

$$\frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma} = 0.$$

The equation of the other tangent at the node is

$$(2\beta\gamma + \alpha^2)\frac{x}{\alpha} + (2\gamma\alpha + \beta^2)\frac{y}{\beta} + (2\alpha\beta + \gamma^2)\frac{z}{\gamma} = 0.$$

13. Considering the critic centre corresponding to a root θ of the cubic equation, the equation of the line joining the other two critic centres is

$$\frac{\lambda x}{\theta + \lambda} + \frac{\mu y}{\theta + \mu} + \frac{\nu z}{\theta + \nu} = 0,$$

which is the polar of the critic centre in regard to the twofold centre conic. The critic centres are consequently conjugate poles in regard to the twofold centre conic.

14. The equation of the tangents at the critic centre considered as a node of the corresponding cubic curve is

$$\big(\theta + 4\lambda,\ \theta + 4\mu,\ \theta + 4\nu,\ -\theta - \tfrac{2\mu\nu}{\theta},\ -\theta - \tfrac{2\nu\lambda}{\theta},\ -\theta - \tfrac{2\lambda\mu}{\theta}\big)(x, y, z)^2 = 0.$$

15. The last-mentioned formulae lead to some which involve the three critic centres viz. if $X = 0$, $Y = 0$, $Z = 0$ are the equations of the sides of the triangle formed by the critic centres, then the equations of the tangents at the three critic centres respectively are of the form

$$BY^2 + CZ^2 = 0, \qquad AX^2 + CZ^2 = 0, \qquad AX^2 + BY^2 = 0,$$

so that the tangents in question are in fact the tangents from the three nodes respectively to the conic

$$AX^2 + BY^2 + CZ^2 = 0 :$$

the three nodes or critic centres being thus conjugate poles in regard to the conic; this is called "the three-centre conic."

16. The equation of a nodal cubic is also expressible in a simple form in terms of the new coordinates X, Y, Z. In the formulae for these transformations, and indeed throughout the memoir, the three roots of the equation in θ are represented by $\theta_1, \theta_2, \theta_3$, and I write also

$$l_1 = \theta_2 - \theta_3, \qquad l_2 = \theta_3 - \theta_1, \qquad l_3 = \theta_1 - \theta_2,$$

$$e_i = (\theta_i + \lambda)(\theta_i + \mu)(\theta_i + \nu) \quad (i = 1, 2, 3).$$

17. If $\lambda a + \mu b + \nu c = 0$, that is, if (a, b, c) are the coordinates of a point on the line $\lambda x + \mu y + \nu z = 0$, then the critic centres lie on the cubic

$$\frac{a}{x} + \frac{b}{y} + \frac{c}{z} - \frac{2(a+b+c)}{x+y+z} = 0,$$

or, what is the same thing, this curve is the locus of the critic centres corresponding to the several lines $\lambda x + \mu y + \nu z = 0$ through the point (a, b, c).

18. In particular, taking in succession for the point (a, b, c) the point of intersection of the line $\lambda x + \mu y + \nu z = 0$ with the lines $x = 0$, $y = 0$, $z = 0$, $x + y + z = 0$, the critic centres lie on the conics

$$\frac{\mu}{y} + \frac{\nu}{z} - \frac{2(\mu+\nu)}{x+y+z} = 0,$$
$$\frac{\nu}{z} + \frac{\lambda}{x} - \frac{2(\nu+\lambda)}{x+y+z} = 0,$$
$$\frac{\lambda}{x} + \frac{\mu}{y} - \frac{2(\lambda+\mu)}{x+y+z} = 0,$$
$$\frac{\lambda}{x} + \frac{\mu}{y} + \frac{\nu}{z} = 0,$$

which are useful for the construction of the critic centres for a given line $\lambda x + \mu y + \nu z = 0$. The last of the four conics passes through the point $(1, 1, 1)$, which is the harmonic of the line $x + y + z = 0$ in regard to the triangle formed by the lines $x = 0$, $y = 0$, $z = 0$; and I call it the *harmonic conic.*

Article Nos. 19 to 21. General Formulae for the Critic Centres.

19. I consider the involution

$$xyz + k(x+y+z)^2(\lambda x + \mu y + \nu z) = 0.$$

Writing the equation in the form

$$k(\lambda x + \mu y + \nu z) = \frac{-xyz}{(x+y+z)^2},$$

and differentiating with regard to x, y, z respectively, we obtain

$$\begin{aligned} -k\lambda(x+y+z)^3 &= yz(-x+y+z), \\ -k\mu(x+y+z)^3 &= zx(\ \ x-y+z), \\ -k\nu(x+y+z)^3 &= xy(\ \ x+y-z), \end{aligned}$$

which determine the coordinate ratios $x : y : z$ of the node or critic centre; and the corresponding value of k.

20. Writing the equations under the form

$$\frac{-k(x+y+z)^3}{xyz} = \frac{-x+y+z}{\lambda x} = \frac{x-y+z}{\mu y} = \frac{x+y-z}{\nu z} = \frac{2}{\theta},$$

where θ is an auxiliary parameter to be determined, we find

$$x\Big(1 + \tfrac{2\lambda}{\theta}\Big) + y + z = 2x\Big(1 + \tfrac{\lambda}{\theta}\Big),$$

that is $x+y+z = 2x\left(1+\frac{\lambda}{\theta}\right)$, and consequently

$$x+y+z \;=\; 2x\left(1+\tfrac{\lambda}{\theta}\right) \;=\; 2y\left(1+\tfrac{\mu}{\theta}\right) \;=\; 2z\left(1+\tfrac{\nu}{\theta}\right),$$

and we have thence an equation which may also be written in the form

$$\frac{1}{\theta+\lambda}+\frac{1}{\theta+\mu}+\frac{1}{\theta+\nu}-\frac{1}{\theta} \;=\; 0,$$

that is,

$$\theta^3-\theta(\mu\nu+\nu\lambda+\lambda\mu)-2\lambda\mu\nu \;=\; 0;$$

and we then have

$$k \;=\; -\frac{\frac{1}{2}\theta^3}{(\theta+\lambda)(\theta+\mu)(\theta+\nu)}.$$

21. We see that θ is determined by a cubic equation, and that the ratios $x:y:z$ and the parameter k are rational functions of θ. There are thus three nodes or critic centres, and the like number of nodal curves and of critic values of k.

The form secondly obtained for the equation in θ shows that we may write

$$x:y:z:x+y+z:\lambda x+\mu y+\nu z \;=\; \frac{1}{\theta+\lambda}:\frac{1}{\theta+\mu}:\frac{1}{\theta+\nu}:\frac{1}{\theta}:1.$$

Article Nos. 22 to 32, relating to a Twofold and a One-with-Twofold Centre.

22. If k has a twofold and a one-with-twofold value, then θ will have also a twofold and a one-with-twofold value; and conversely. The equation in θ will have a twofold and a one-with-twofold root if

$$(\mu\nu+\nu\lambda+\lambda\mu)^3-27\,\lambda^2\mu^2\nu^2=0;$$

or, what is the same thing, if

$$\mu\nu+\nu\lambda+\lambda\mu-3(\lambda\mu\nu)^{2/3} \;=\; 0,$$

or finally if

$$\sqrt{\lambda}+\sqrt{\mu}+\sqrt{\nu} \;=\; 0,$$

so that the condition is satisfied if $\lambda=\alpha^{-2}$, $\mu=\beta^{-2}$, $\nu=\gamma^{-2}$ where $\alpha+\beta+\gamma=0$. In fact with these values the equation in θ becomes

$$(\alpha\beta\gamma\,\theta)^3-3\alpha\beta\gamma\,\theta-2=0,$$

that is

$$(\alpha\beta\gamma\,\theta+1)^2(\alpha\beta\gamma\,\theta-2)=0,$$

so that the twofold value is $\theta_1=\theta_2=-\frac{1}{\alpha\beta\gamma}$; and the one-with-twofold value is $\theta_3=\frac{2}{\alpha\beta\gamma}$.

23. It is throughout assumed that the quantities α,β,γ satisfy the condition $\alpha+\beta+\gamma=0$. The result just obtained shows that the line

$$\frac{x}{\alpha^2}+\frac{y}{\beta^2}+\frac{z}{\gamma^2}=0$$

is a twofold and one-with-twofold satellite line. From this equation, considering α,β,γ as variable parameters satisfying the condition $\alpha+\beta+\gamma=0$, we find at once the equation of the curve enveloped by the line in question, which curve is called simply the envelope—viz. the coordinates of the point of contact are found to be $x : y : z = \alpha^4 : \beta^4 : \gamma^4$, and thence the equation of the envelope is $\sqrt{x} + \sqrt{y} + \sqrt{z} = 0$, or rationalising, it is

$$x^4+y^4+z^4-4(yz^3+y^3z+zx^3+z^3x+xy^3+x^3y)+6(y^2z^2+z^2x^2+x^2y^2)-124(x^2yz+y^2zx+z^2xy) = 0.$$

The before-mentioned equation $\lambda^{-1/2} + \mu^{-1/2} + \nu^{-1/2} = 0$, or

$$(\mu\nu + \nu\lambda + \lambda\mu)^3 - 27\,\lambda^2\mu^2\nu^2 = 0,$$

may be considered as the tangential equation; the envelope is thus of the order 4, and the class 6.

24. It is easy to show that the curve has three nodes the coordinates whereof are $(-4, 1, 1)$, $(1, -4, 1)$, $(1, 1, -4)$; and this being known, the equation may be transformed so as to put the nodes in evidence. I effect the transformation synthetically as follows, viz. writing $x+y+z = \sigma$, $yz + zx + xy = q$, $xyz = r$, the equation of the curve is

$$(\sigma^4 - 2q\sigma^2 + 2q^2 + 4r\sigma) - 4(q\sigma^2 - 2q^2 - r\sigma) + 6(q^2 - 2r\sigma) - 124(r\sigma) = 0,$$

viz. it is

$$\sigma^4 - 6q\sigma^2 + 16q^2 - 128r\sigma = 0,$$

which is $(\sigma^2 + 4q)^2 - 16\sigma(3\sigma^2 + 4q\sigma + 8r) = 0$, or, putting for a moment $l = -\frac{16}{5}$, and therefore $5(l - 2) = -16$, it is

$$(\sigma^2 + 4q)^2 + (l - 2)\,5\sigma(3\sigma^2 + 4q\sigma + 8r) = 0.$$

Now writing $x' = \sigma + 2x = 3x + y + z$, $y' = \sigma + 2y = x + 3y + z$, $z' = \sigma + 2z = x + y + 3z$, we find

$$y'z' + z'x' + x'y' = 7\sigma^2 + 4q, \qquad x' + y' + z' = 5\sigma, \qquad x'y'z' = 3\sigma^3 + 4q\sigma + 8r,$$

so that the equation is

$$(y'z' + z'x' + x'y')^2 + (l - 2)\,x'y'z'(x' + y' + z') = 0,$$

that is

$$y'^2z'^2 + z'^2x'^2 + x'^2y'^2 + l\,x'y'z'(x' + y' + z') = 0;$$

or, putting for l its value, the equation is

$$5(y'^2z'^2 + z'^2x'^2 + x'^2y'^2) - 16\,x'y'z'(x' + y' + z') = 0;$$

or, as this may also be written,

$$(5,\ 5,\ 5,\ -3,\ -3,\ -3 \mid y'z',\ z'x',\ x'y')^2 = 0;$$

a form which shows that the curve has three nodes at the angles of the triangle $x' = 0$, $y' = 0$, $z' = 0$.

25. It is easy to see that the curve is touched by the lines $x = 0$, $y = 0$, $z = 0$ at their intersections with the lines $y - z = 0$, $z - x = 0$, $x - y = 0$ respectively, or (what is the same thing) in the points $(0, 1, 1)$, $(1, 0, 1)$, $(1, 1, 0)$ respectively. It may be added that the line $y - z = 0$ meets the curve in the node $(-4, 1, 1)$, being of course a point of twofold intersection, in the point $(0, 1, 1)$ on the line $x = 0$, and besides in the point $(16, 1, 1)$: and the like for the lines $z - x = 0$ and $x - y = 0$.

26. It may be noticed that although any line passing through one of the nodes is in a sense a tangent to the envelope, yet that it is not a proper tangent and does not give rise to a twofold centre. It is in fact shown (*post*, Nos. 73 and 74) that the critic centres for a line $\lambda x+\mu y+\nu z=0$ passing through the point $(-4,1,1)$ are three points lying, one of them on the line $y+z=0$, and the other two on the conic $x(x+y+z)-6yz=0$.

27. Assume that θ has its twofold value $=-\frac{1}{\alpha\beta\gamma}$; the equations

$$x:y:z=\frac{1}{\theta+\lambda}:\frac{1}{\theta+\mu}:\frac{1}{\theta+\nu},$$

substituting therein for λ,μ,ν the values $\alpha^{-2},\beta^{-2},\gamma^{-2}$, give for the coordinates of the twofold centre

$$x:y:z=\frac{1}{\beta\gamma-\alpha^2}:\frac{1}{\gamma\alpha-\beta^2}:\frac{1}{\alpha\beta-\gamma^2};$$

but in virtue of the relation $\alpha+\beta+\gamma=0$ we have

$$\beta\gamma-\alpha^2=\beta\gamma+\gamma\alpha+\alpha\beta=\gamma\alpha-\beta^2=\alpha\beta-\gamma^2;$$

or the values are $x:y:z=\alpha^3:\beta^3:\gamma^3$. Hence also we have as the equation of the locus of the twofold centre, $\sqrt{x}+\sqrt{y}+\sqrt{z}=0$, or, what is the same thing,

$$x^2+y^2+z^2-2yz-2zx-2xy=0,$$

which is a conic touching the lines $x=0$, $y=0$, $z=0$ at their intersections with the lines $y-z=0$, $z-x=0$, $x-y=0$ respectively, or, what is the same thing, in the points $(0,1,1)$, $(1,0,1)$, $(1,1,0)$ respectively.

28. Similarly, if θ has its one-with-twofold value $=\frac{2}{\alpha\beta\gamma}$, the equations

$$x:y:z=\frac{1}{\theta+\lambda}:\frac{1}{\theta+\mu}:\frac{1}{\theta+\nu},$$

substituting therein for λ,μ,ν the values $\alpha^{-2},\beta^{-2},\gamma^{-2}$, give for the coordinates of the one-with-twofold centre

$$x:y:z=\frac{1}{2\alpha^2+\beta\gamma}:\frac{1}{2\beta^2+\gamma\alpha}:\frac{1}{2\gamma^2+\alpha\beta};$$

but in virtue of $\alpha+\beta+\gamma=0$ we have

$$2\alpha^2+\beta\gamma=\alpha^2-\alpha(\beta+\gamma)+\beta\gamma=(\alpha-\beta)(\alpha-\gamma)=-(\gamma-\alpha)(\alpha-\beta),$$

and similarly

$$2\beta^2+\gamma\alpha=-(\alpha-\beta)(\beta-\gamma),\qquad 2\gamma^2+\alpha\beta=-(\beta-\gamma)(\gamma-\alpha);$$

and thence these values are

$$x:y:z=\alpha^2(\beta-\gamma):\beta^2(\gamma-\alpha):\gamma^2(\alpha-\beta),$$

for the coordinates of the one-with-twofold centre.

We thence deduce

$$y+z=\beta^2\gamma-\beta^3+\gamma^2\alpha-\gamma^3\beta=(\beta\gamma-\alpha\beta-\alpha\gamma)(\beta-\gamma)=(\beta\gamma+\alpha^2)(\beta-\gamma),$$

and consequently $-x+y+z=\beta\gamma(\beta-\gamma)$, that is

$$x:y:z:-x+y+z:x-y+z:x+y-z=\alpha^2(\beta-\gamma):\beta^2(\gamma-\alpha):\gamma^2(\alpha-\beta):\beta\gamma(\beta-\gamma):\gamma\alpha(\gamma-\alpha):\alpha\beta(\alpha-\beta),$$

and these give

$$-(-x+y+z)(x-y+z)(x+y-z)+xyz=0,$$

or, what is the same thing,

$$x^3+y^3+z^3-(yz^2+y^2z+zx^2+z^2x+xy^2+x^2y)+3xyz=0,$$

as the equation of the locus of the one-with-twofold centre, which locus is thus a cubic curve.

29. The equation

$$x^3+y^3+z^3-(yz^2+y^2z+zx^2+z^2x+xy^2+x^2y)+3xyz=0$$

of the one-with-twofold centre locus may be transformed as follows, viz. writing for a moment $x+y+z=-w$, we have

$$\begin{aligned}(9x+4w)(9y+4w)(9z+4w)-w^3 &= 729\,xyz+324\,w(yz+zx+xy)-144w^2+64w^3-w^3,\\ &= 81\{9xyz+4w(yz+zx+xy)-w^3\},\\ &= 81\left[9xyz-12xyz-4(yz^2+\&\text{c.})+(x^3+y^3+z^3)-(3yz^2+\&\text{c.})+6xyz\right],\\ &= 81\{x^3+y^3+z^3-(yz^2+\&\text{c.})+3xyz\},\end{aligned}$$

so that the equation may be written

$$(9x+4w)(9y+4w)(9z+4w)-w^3=0,$$

or, what is the same thing,

$$(5x-4y-4z)(-4x+5y-4z)(-4x-4y+5z)+(x+y+z)^3=0,$$

which shows that the intersections of the line $x+y+z=0$ with the sides $x=0$, $y=0$, $z=0$ of the triangle are inflexions on the curve; and that the tangents at these points are respectively $5x-4y-4z=0$, $-4x+5y-4z=0$, $-4x-4y+5z=0$.

30. The curve passes through the point $(1,1,1)$, which is the harmonic of $x+y+z=0$ in regard to the triangle; and this point is moreover a node on the curve; in fact if the equation be represented by $W=0$, then we have

$$\partial_x W=3x^2-2x(y+z)-y^2+3yz-z^2,$$

$=0$ for the point in question; and similarly $\partial_y W=0$, and $\partial_z W=0$.

31. The equation for the twofold centre conic may also be obtained as follows; viz. the equations

$$\frac{-x+y+z}{\lambda x}=\frac{x-y+z}{\mu y}=\frac{x+y-z}{\nu z}=\frac{2}{\theta},$$

give

$$(-x+y+z)(x-y+z)(x+y-z)=\tfrac{8}{\theta^3}\,xyz\,\lambda\mu\nu,$$

which substituting for θ the twofold value $=-\frac{1}{\alpha\beta\gamma}$, gives

$$(-x+y+z)(x-y+z)(x+y-z)+8xyz=0,$$

an equation which may be written

$$(x+y+z)^2(x^2+y^2+z^2-2yz-2zx-2xy)=0,$$

which is the former result affected by the extraneous factor $x+y+z$.

If, instead, we substitute for θ the onefold value $=\frac{2}{\alpha\beta\gamma}$, we find

$$-(-x+y+z)(x-y+z)(x+y-z)+xyz=0,$$

or, what is the same thing,

$$x^3+y^3+z^3-(yz^2+y^2z+zx^2+z^2x+xy^2+x^2y)+3xyz=0,$$

which is the one-with-twofold centre cubic.

32. Recollecting that

$$k=-\frac{\frac{1}{2}\theta^3}{(\theta+\lambda)(\theta+\mu)(\theta+\nu)},$$

we deduce for the twofold value of k

$$k_1=k_2=-\frac{2\alpha^2\beta^2\gamma^2}{(\beta\gamma+\gamma\alpha+\alpha\beta)^3},$$

and for the one-with-twofold value

$$k_3=\frac{4\alpha\beta\gamma}{(2\alpha^2+\beta\gamma)(2\beta^2+\gamma\alpha)(2\gamma^2+\alpha\beta)}=-\frac{4\alpha\beta\gamma}{(\beta-\gamma)^2(\gamma-\alpha)^2(\alpha-\beta)^2}.$$

Article Nos. 33 to 38, relating to the Tangents at a Node or Critic Centre.

33. I proceed to investigate the equation of the tangents at the node of the curve

$$xyz+k(x+y+z)^2(\lambda x+\mu y+\nu z)=0;$$

it will be recollected that if x,y,z are the coordinates of the node, then we have

$$x:y:z:x+y+z:\lambda x+\mu y+\nu z=\frac{1}{\theta+\lambda}:\frac{1}{\theta+\mu}:\frac{1}{\theta+\nu}:\frac{1}{\theta}:1.$$

Representing for a moment the equation of the curve by $U=0$, then the second derived functions of U are

$$k\cdot 2(\lambda x+\mu y+\nu z)+4k(x+y+z)\lambda,$$
$$k\cdot 2(\lambda x+\mu y+\nu z)+4k(x+y+z)\mu,$$
$$k\cdot 2(\lambda x+\mu y+\nu z)+4k(x+y+z)\nu,$$
$$x+k\cdot 2(\lambda x+\mu y+\nu z)+2k(x+y+z)(\mu+\nu),$$
$$y+k\cdot 2(\lambda x+\mu y+\nu z)+2k(x+y+z)(\nu+\lambda),$$
$$z+k\cdot 2(\lambda x+\mu y+\nu z)+2k(x+y+z)(\lambda+\mu),$$

or calling these (a,b,c,f,g,h) respectively, and substituting the values $x=\frac{1}{\theta+\lambda}$, &c., we find

$$a=2k+\frac{4k\lambda}{\theta}=\frac{2k}{\theta}(\theta+4\lambda),$$

with the like values for b, c; and

$$f=\frac{1}{\theta+\lambda}+\frac{2k}{\theta}(\theta+2(\mu+\nu)),$$

where the term in $(\ldots)$ is, after computation,

$$=-2\theta-2\mu-2\nu-\tfrac{2\mu\nu}{\theta}+\theta+2\mu+2\nu=-\theta-\tfrac{2\mu\nu}{\theta};$$

$$\text{i.e. } f=\frac{2k}{\theta}\left(-\theta-\tfrac{2\mu\nu}{\theta}\right),$$

with the like values for g and h; or omitting the common factor $\frac{2k}{\theta}$, we have

$$(a, b, c, f, g, h) = \big(\theta + 4\lambda,\ \theta + 4\mu,\ \theta + 4\nu,\ -\theta - \tfrac{2\mu\nu}{\theta},\ -\theta - \tfrac{2\nu\lambda}{\theta},\ -\theta - \tfrac{2\lambda\mu}{\theta}\big),$$

and thence, taking now x, y, z as current coordinates, the equation of the tangents at the node is

$$(a, b, c, f, g, h \mid x, y, z)^2 = 0.$$

34. Substituting for λ, μ, ν the values $\alpha^{-2}, \beta^{-2}, \gamma^{-2}$, and for θ the twofold value $-\frac{1}{\alpha\beta\gamma}$, the equation of the tangents at the twofold centre becomes

$$\big(\beta^2\gamma^2(4\beta\gamma - \alpha^2),\ \ldots,\ \alpha^2\beta\gamma(\beta\gamma + 2\alpha^2),\ \ldots\big)(x, y, z)^2 = 0,$$

which is at once reduced to

$$\big(\beta^2\gamma^2(\beta - \gamma)^2,\ \ldots,\ \alpha^2\beta\gamma(\gamma - \alpha)(\alpha - \beta),\ \ldots\big)(x, y, z)^2 = 0,$$

or, what is the same thing,

$$\big[\beta\gamma(\beta - \gamma)x + \gamma\alpha(\gamma - \alpha)y + \alpha\beta(\alpha - \beta)z\big]^2 = 0,$$

which shows that the twofold centre is a cusp, and that the tangent is

$$\beta\gamma(\beta - \gamma)x + \gamma\alpha(\gamma - \alpha)y + \alpha\beta(\alpha - \beta)z = 0,$$

or, what is the same thing,

$$(\beta - \gamma)\frac{x}{\alpha} + (\gamma - \alpha)\frac{y}{\beta} + (\alpha - \beta)\frac{z}{\gamma} = 0.$$

35. Writing in like manner $\lambda, \mu, \nu = \alpha^{-2}, \beta^{-2}, \gamma^{-2}$ and θ for the one-with-twofold value $= \frac{2}{\alpha\beta\gamma}$, we find for the equation of the tangents at the one-with-twofold centre

$$\big(2\beta^2\gamma^2(2\beta\gamma + \alpha^2),\ \ldots,\ -\alpha^2\beta\gamma(4\beta\gamma + \alpha^2),\ \ldots\big)(x, y, z)^2 = 0,$$

which may be reduced to

$$\big(2\beta^2\gamma^2(2\beta\gamma + \alpha^2),\ \ldots,\ \alpha^2\beta\gamma(2\gamma\alpha + \beta^2)(2\alpha\beta + \gamma^2),\ \ldots\big)(x, y, z)^2 = 0,$$

or, what is the same thing,

$$(\beta\gamma x + \gamma\alpha y + \alpha\beta z)\big[(2\beta\gamma + \alpha^2)\beta\gamma x + (2\gamma\alpha + \beta^2)\gamma\alpha y + (2\alpha\beta + \gamma^2)\alpha\beta z\big] = 0.$$

Hence at the one-with-twofold centre the equation of one of the tangents is

$$(2\beta\gamma + \alpha^2)\beta\gamma x + (2\gamma\alpha + \beta^2)\gamma\alpha y + (2\alpha\beta + \gamma^2)\alpha\beta z = 0,$$

or, as this may otherwise be written,

$$(2\beta\gamma + \alpha^2)\frac{x}{\alpha} + (2\gamma\alpha + \beta^2)\frac{y}{\beta} + (2\alpha\beta + \gamma^2)\frac{z}{\gamma} = 0.$$

The memoir of Paper 349 continues from Art. 36 with the discussion of the harmonic-conic transformation, the three-centre conic, formulae for X, Y, Z, and the general formulae for arbitrary critic centres. The remainder of the paper, comprising Arts. 36–80 (facsimile pp. 326–354), falls beyond the present 50-page extract and will be taken up in the subsequent installment.

End of excerpt: PDF pp. 301–350 of Cayley, Coll. Math. Papers, Vol. V.